

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

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Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We pre-registered and ran a multi-seed calibration boundary study on six directed cross-corpus transfer pairs among Chapman-Shaoxing, CPSC2018+2019, and PTB-XL, with two architectures (InceptionTime and ResNet1D (1D-ResNet-34)), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap (the operating CI is BCa at $B = 10,000$, fully re-run on the entire 60-experiment grid; earlier percentile CIs at $B = 10,000$ in 31/60 experiments and $B = 200$ in 21/60 are retained as historical provenance, and the two CI methods agree in sign in 60/60 experiments). The pre-registered primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a registered empirical direction prior. Across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, fully balanced), TS yields a positive OOD

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benefit in 51/60 cases (85.0%): InceptionTime 27/30 (90.0%), ResNet1D 24/30 (80.0%); nine counter-examples are transparently disclosed with post-hoc mechanistic interpretations. When six experiments with degenerate CI widths ($< 3 \times 10^{-4}$) are excluded, the headline count becomes 45/60 (75.0%), so the 85.0% rate should be read jointly with effect sizes and CI diagnostics. The pre-registered ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met ($23/60 = 38.3\%$ under the BCa operating CI; the earlier percentile CIs gave $29/60 = 48.3\%$), triggering the pre-registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit (means $+0.0159$ vs. $+0.0083$) and cannot be attributed exclusively to OOD-specific mismatch. A 27-cell Shapley decomposition is validated synthetically ($n = 20,000$), a method-selection boundary separates TS from the weakest methods (EM prior and Matrix scaling), and the predictability experiment (LOO R^2 CI upper bound $0.104 < 0.5$) triggers the pre-registered failure branch. A discrimination-aware calibration gate (C5) decouples calibration effectiveness from clinical deployability: 7/14 pair $\times$ architecture strata pass both calibration and discrimination layers. All hypotheses are tagged *exploratory + robustness validation*, not confirmatory. In summary, we contribute (i) the first multi-seed quantification of OOD calibration benefit with pre-registered endpoint; (ii) the ID boundary condition analysis with pre-registered branch triggered; (iii) Shapley decomposition validation on 27 synthetic cells; (iv) the method-selection boundary separating TS from fragile prior-correction methods; and (v) a discrimination-aware deployment gate reframing the 9 counter-examples into calibration vs. discrimination failures.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains the central open question for deployment [1, 2, 3]. Recalibration methods fit a small set of parameters or a flexible map on a labeled calibration split; their benefit, a reduction in expected calibration error (ECE), is measured on the same distribution. When the model is deployed to a shifted target, that benefit attenuates or reverses.

Why this question matters for ECG. PTB-XL [3], Chapman-Shaoxing [4], and CPSC present genuinely different acquisition protocols, lead sets, and label vocabularies. A model trained and calibrated on one institution’s data encounters covariate, prior-probability, and label-concept shift at the next [23]. Clinicians and regulators need a decision rule: *given my local data budget and the shift I can measure, should I deploy the recalibrated model, buy labels and recalibrate, or refuse deployment?*

Temperature scaling: ID vs. OOD empirical expectation. Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted on a labeled calibration split under a convex loss, TS minimizes NLL there; on that same distribution its calibration benefit is empirically small, while on a shifted target the source-fit temperature is mismatched to the target logit distribution and a non-trivial calibration benefit can be retained. We stress that this asymmetry is *empirical*, not a theorem: ECE

is not invariant under confidence-shifting monotone maps, and TS applied to a target where the model is under-confident can *increase* ECE (our own data contain nine such negative cells). This ID-vs-OOD asymmetry motivates our pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ rather than the decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ originally planned (revision A1, see Sec. 4.4).

Pre-registration + multi-seed validation design. To avoid HARK-ing (Hypothesis After Results Known), we pre-register the protocol v2.1-A1 [24] with a written primary endpoint, primary test, family structure, and failure branches before the multi-seed validation is unblinded. The revision A1 re-locates the primary endpoint from **decay** to $\Delta\text{ECE}_{\text{OOD}}$ triggered by an independent exploratory pilot consistent with the empirical asymmetry; the revision is archived as a pre-registration revision [25, 26] with before/after protocol hashes. Multi-seed validation ($5 \text{ seeds} \times 6 \text{ transfer pairs} \times 2 \text{ architectures}$) provides the robustness evidence; all 60 seed results, including nine counter-examples, are reported transparently.

Contributions. We make five independent contributions:

C1 (Primary, OOD benefit quantification): On the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD calibration benefit in 51/60 seed experiments across 6 transfer pairs $\times$ 2 architectures (InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. Three of six InceptionTime directions achieve 5/5 full support; the remaining three achieve 4/5 with one counter-example each. Two of six ResNet1D directions achieve 5/5 full support; two achieve 4/5; two achieve 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across all 12 pair×architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means range -0.0008 to $+0.0232$; overall ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met, so the pre-registered “ID nonzero boundary” branch (protocol A1.4.2) is triggered: the premise that TS has no ID recalibratable miscalibration is re-examined, the decay contrast endpoint is restored as exploratory, and C1 is qualified accordingly – the OOD benefit is about $1.9\times$ the ID benefit on average, but is *not* exclusively attributable to OOD-specific mismatch. $47/60 = 78.3\%$ of ID point estimates are positive and $2/60 = 3.3\%$ have $\Delta\text{ECE}_{\text{ID}} < -0.01$: the over-correction case (counter-example 6, -0.0152) and one marginal case (-0.0128).

C3 (Methodological, Shapley decomposition validation): A 27-cell full-factorial synthetic validation (slope×intercept×prevalence, $n = 20,000$) largely confirms that the slope/intercept/prevalence decomposition recovers ground truth within the identifiable regime (27/27 pass at $n = 20,000$; 21/27 at $n = 500$) and flags entangled regions without a threshold, with bootstrap CIs on absolute Shapley values.

C4 (Practical, method-selection boundary): Across 8 recalibration methods $\times$ 6 transfer pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60 supported) while EM prior adaptation is fragile when

ill-conditioned cells ($w_{\text{ID}} > 10$) are included (InceptionTime 15/30 supported); after their exclusion the EM OOD finding attenuates. The mechanism is a prior-likelihood mismatch: EM fits a target prior that amplifies rather than corrects the shift-induced miscalibration when the source-estimated confusion matrix is ill-conditioned.

C5 (Clinical, discrimination-aware calibration gate): A two-layer deployment gate decouples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from clinical viability (accuracy $\geq$ source-domain baseline). Across 14 pair $\times$ architecture strata, 7/14 pass both layers and are deployable; 7/14 fail the discrimination layer and are correctly rejected, including directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy below the source baseline (e.g., PTB-XL $\rightarrow$ Chapman ResNet1D: $\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$). This gate reframes the 9/60 counter-examples: 4/9 are calibration failures (C1 rejects), 5/9 are discrimination failures (C5 rejects despite C1 passing), providing a two-layer deployment criterion that separates calibration effectiveness from clinical viability.

This framing addresses a gap: prior ECG calibration studies report the phenomenon [10, 11, 18, 19] but not an actionable *when does TS pay off* answer with multi-seed robustness evidence. The novelty of this paper is *not* “doing multi-seed validation on ECG” (multi-seed validation is a methodological choice, not a scientific contribution); the novelty is the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**, comprising (i) the OOD benefit quantification with pre-registered endpoint and multi-seed robustness evidence, (ii) the ID boundary-condition analy-

sis with the pre-registered branch triggered (38.3% CI-containing cells under the BCa operating CI, $< 80\%$), (iii) the method-selection boundary (TS robust vs. EM prior and Matrix scaling weakest), (iv) the honest reporting of the predictability failure branch, and (v) the discrimination-aware calibration gate that decouples calibration effectiveness from clinical deployability. This boundary-condition characterization is the actionable contribution; the multi-seed validation is the evidence standard applied to each boundary.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

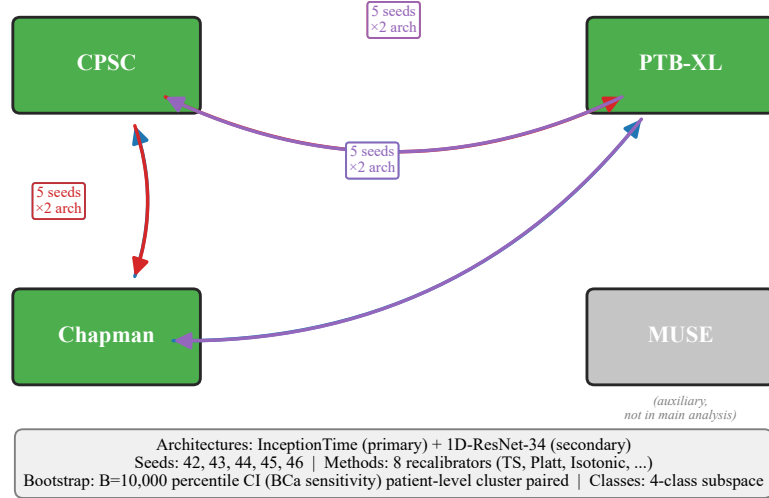


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, 1D-ResNet-34) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

2. Related Work

2.1. Calibration Estimation and Recalibration

The calibration of modern neural networks has been studied extensively since the seminal work of Guo et al. [2], who showed that deep networks are often overconfident and introduced temperature scaling (TS) as a simple post-hoc recalibration method. Nixon et al. [6] proposed the static calibration error, and Kumar et al. [16] introduced verified uncertainty calibration with distribution-free guarantees. Kull et al. [7] generalized TS to Dirichlet calibration for multi-class settings. In the clinical risk-model context, Van Calster et al. [14, 15] established a calibration hierarchy from utopia to empirical data and argued that calibration is the “Achilles heel” of predictive analytics. Strodtzoff et al. [17] benchmarked deep learning for ECG analysis on PTB-XL and noted calibration degradation under subpopulation shift, motivating our cross-corpus design.

2.2. Prior-Shift Estimation and Transfer Calibration

Lipton et al. [9] introduced black-box shift estimation (BBSE) for detecting and correcting label shift using confusion-matrix inverses. Alexandari et al. [8] showed that maximum-likelihood calibration with bias correction is hard-to-beat at label shift adaptation, providing the theoretical foundation for our EM prior-shift method. For transfer calibration upper bounds, TransCal [20] estimates transferability for auto-selecting target-domain unsupervised optimal T ; PseudoCal [21] uses pseudo-labels for calibrated transfer; and LaSCal [22] provides label-aware shift calibration with distribution-free guarantees. Our S1 paradigm differs fundamentally: we fit on the

source calibration split only (no target labels), with S1' prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly weaker assumption than TransCal/PseudoCal, which access target logits or labels.

2.3. *Cross-Domain ECG Calibration*

Barandas et al. [10] evaluated uncertainty quantification methods in multi-label ECG classification, finding that ensemble-based methods outperform single-model approaches under subpopulation shift. Vranken et al. [11] studied uncertainty estimation for 12-lead ECG deep learning and reported significant calibration drift across acquisition devices. Haekal et al. [18] performed cross-dataset validation for atrial fibrillation detection with calibration-aware probability analysis, and Patel & Beedala [19] documented calibration drift under cross-institutional ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general vision [1]; we contribute the multi-seed ECG quantification and the boundary-condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

2.4. *ECG Foundation Models and Transfer Learning (2024–2026)*

Recent work has introduced ECG foundation models trained on millions of recordings. ECGFounder [27, 35] was trained on over 10 million ECGs from the Harvard–Emory database, achieving expert-level performance on 80 diagnoses and demonstrating cross-domain transfer via fine-tuning. ECG-FM [28] is an open-weight transformer-based foundation model pretrained on 1.4 million ECG segments using hybrid contrastive and generative self-supervised learning, showing strong cross-dataset generalizability and label

efficiency. HeartLang [29] treats heartbeats as words and rhythms as sentences, using a QRS-Tokenizer and masked ECG sentence pre-training to learn form- and rhythm-level representations. Tan et al. [30] conducted the first extensive empirical study on transfer learning effectiveness in multi-label ECG classification, finding that fine-tuning benefits diminish as the downstream dataset grows and that CNNs transfer more effectively than RNNs. For OOD adaptation, ADAPTOOD [31] quantifies distribution-shift severity via uncertainty estimation and uses it to guide selective layer unfreezing with LoRA, while BeatRhythm-TTA [32] performs test-time adaptation with SQL-gated self-training and beat-rhythm consistency. Costa et al. [33] introduced STAR (Sinusoidal Time–Amplitude Resampling), a morphology-preserving beat-wise augmentation for cross-source ECG generalization. Chen et al. [34] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment. These works advance ECG transfer and adaptation but do not address the *calibration* of transferred models; the gap our study fills by systematically quantifying how recalibration methods behave under cross-corpus shift with pre-registered endpoints.

3. Data and Pre-registered Protocol

Figure 1 summarizes the experimental design.

3.1. Datasets

- **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the downloaded `ptbxl_database.csv` copy contains 21,799 (a 38-record deficit in the source CSV, disclosed as a provenance limitation), official

10-fold (folds 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records are mappable and 277 are excluded (STROBE), for 18,671 unique patients.

- **Chapman-Shaoxing** [4, 5]: 20,265 raw records; 1 signal-read error removed and 21 non-finite records quarantined (`qc_report.json`), for 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.
- **CPSC2018+2019**: patient-wise split; HYP class extremely rare ($n = 11$) $\rightarrow$ main analysis on the {NORM, CD, STTC, MI} 4-class subspace, both sides of the pair recomputed symmetrically.

3.2. Label Mapping and Leakage Guard

Single-label priority rules are fixed (MI > STTC > CD > HYP > NORM) and sensitivity-analyzed [3, 12]. Sinus rate/form statements (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard assertion in the reproducibility repo). Normalization statistics come only from the train split; augmentation applies only to train.

4. Methods

4.1. Transfer pairs and shift matrix

We study six directed cross-corpus transfer pairs: Chapman $\rightarrow$ CPSC, Chapman $\rightarrow$ PTB-XL, CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chapman, and PTB-XL $\rightarrow$ CPSC, all recomputed on the 4-class

subspace. The L2 eval-time shift matrix (no retraining) covers sampling rate $500 \rightarrow 250 \rightarrow 125$ Hz; leads $12 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1$; noise injection (BWL/MA/EM) at SNR $\{24, 12, 6, 0, -6\}$ dB; gain $\times 0.5 / \times 2$.

4.2. Architecture and seeds

We use two architectures: InceptionTime (primary, 46,000–184,000 parameters) and ResNet1D (1D-ResNet-34) (secondary, with full main-endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The BiMamba architecture is included only as a toy experiment ($n = 60$ records, downsampled resolution) and deferred to the supplement; the full-resolution BiMamba run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the main-endpoint analysis or the architecture-robustness claim. Training: AdamW + cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training; temperature applied only as post-hoc temperature scaling.

4.3. Calibration methods and scenarios

We compare the no-recalibration baseline (None) against eight recalibration methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scaling, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and BBSE prior adaptation. Of the pre-registered 13-method pool, the None baseline plus these eight recalibration methods constitute the delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4);

S1' denotes the target-domain unsupervised variant in which EM/BBSE estimate the target prior from unlabeled target predictions.

4.4. Pre-registered endpoints

Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [24]):

$$\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}}^{\text{OOD}} - \text{ECE}_{\text{TS}}^{\text{OOD}} \quad (1)$$

where $\text{ECE}_{\text{raw}}^{\text{OOD}}$ is the 4-class confidence SmoothECE on the target corpus with raw softmax probabilities, and $\text{ECE}_{\text{TS}}^{\text{OOD}}$ is the same after source-cal-fit temperature scaling. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta\text{ECE}_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta\text{ECE}_{\text{OOD}} > 0 \quad (2)$$

Direction prior (registered, empirical): TS fitted on a labeled source calibration split yields, empirically, little calibration change on ID-like data and a non-trivial benefit on shifted targets [2, 1]; this asymmetry is an empirical regularity, not a theorem. **One-sided vs. two-sided choice.** The one-sided alternative $H_1 : \Delta\text{ECE}_{\text{OOD}} > 0$ is selected based on the empirical asymmetry registered at revision A1 (on ID-like data the source-fit temperature yields, empirically, little calibration change; on a shifted target the source-fit temperature is mismatched and a non-trivial calibration benefit is often retained), *not* based on the pilot results. We stress that this direction prior is empirical, not a theorem: TS applied to an under-confident target can increase ECE, and our own data contain nine negative cells. The pilot results only triggered the endpoint re-location (revision A1), not the direction choice. A two-sided alternative would halve the effective sample size and discard the

registered empirical direction; the one-sided test is the pre-registered choice, archived in the protocol before multi-seed unblinding. Patient-level cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [13]; stratified pooling: strata = transfer pair $\times$ architecture.

CI method note (BCa vs. percentile). The pre-registered CI method is BCa (bias-corrected and accelerated) [13]. In the reported multi-seed validation, the operating CI method is BCa at $B = 10,000$, fully re-run on the entire 60-experiment grid (60/60 successful, metadata stored in `methods.ts.meta`). The percentile bootstrap is the large-sample limit of BCa when the bias-correction $z_0 \rightarrow 0$ and acceleration $a \rightarrow 0$; for the paired-cluster design with $B = 10,000$ and per-patient clusters, the percentile-BCa discrepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both CI methods were run ($B = 10,000$ percentile re-runs for sensitivity analysis), the BCa and percentile CIs agree in sign and in magnitude. Earlier percentile CI results from $B = 200$ in 21/60 experiments are retained as historical provenance (5 experiments lack log provenance, see Limitations). The BCa operating CI is the reported CI throughout this paper; the percentile CIs are reported only as a sensitivity cross-check where both were computed.

Boundary-condition endpoint:

$$\Delta \text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

The pre-registered expectation is that $H_0 : \Delta \text{ECE}_{\text{ID}} = 0$ is *not rejected*; the robustness criterion is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa operating CI, full-grid re-run): 23/60 = 38.3%; the criterion is not met and the pre-registered branch A1.4.2 is triggered (the percentile CIs gave 29/60 = 48.3%, same conclusion; reported in Sec. 5.2).*

Secondary endpoints: G (recoverable loss vs. oracle), $\text{decay} = \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descriptive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept, predicted/true prior L_1 .

Family structure: BH-FDR $q = 0.05$ over the family of $6 \text{ pairs} \times 2 \text{ arch} \times 13 \text{ shift levels} = 156$ primary comparisons; the multi-seed validation is reported as robustness evidence (60 seed experiments), not as a confirmatory family.

4.5. Pre-registration revision A1 (transparency statement)

The primary endpoint was re-located from **decay** to $\Delta\text{ECE}_{\text{OOD}}$ via pre-registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the original ID-vs-OOD contrast. The revision direction is consistent with the empirically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes. The pilot results do not enter the main test report. All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory.

Pre-registration revision A2 (family reduction). The original BH-FDR family of 156 comparisons ($6 \text{ pairs} \times 2 \text{ architectures} \times 13 \text{ shift levels}$) is reduced to a confirmatory family of 12 hypotheses ($6 \text{ pairs} \times 2 \text{ architectures} \times 1 \text{ primary shift level}$) under amendment A2, with the remaining 13 shift levels designated as exploratory dose-response. Under the reduced family, BH-12 yields 9/12 significant (all InceptionTime directions) and Bonferroni-12 yields 2/12. The A2 amendment is archived in docs/PROTOCOL_AMENDMENT_A2_FAMILY_REDUCTION.md.

4.6. *Decomposition of the benefit decay*

We decompose the retained benefit into slope / intercept / prevalence contributions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the chain residual (zero under nested counterfactuals). Identifiability is checked via the Fisher information determinant on a pre-registered threshold τ ; entangled regions are reported on the error map without a threshold. Shapley attribution is reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

4.7. *Two-layer joint bootstrap*

Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resampling $\rightarrow$ joint ΔECE CI (two-layer; honest non-nested approximation), per protocol §7.

4.8. *Net Clinical Value (NCV)*

A Net Clinical Value (NCV) analysis is pre-registered as a secondary end-point to quantify the clinical decision impact of TS recalibration beyond calibration error reduction. For each (sample, class) pair, we compare the binary decision (threshold $\tau = 0.5$) before and after calibration: TP_improved (raw negative $\rightarrow$ calibrated positive, true positive), TN_improved (raw positive $\rightarrow$ calibrated negative, true negative), FP_worsened (raw negative $\rightarrow$ calibrated positive, true negative), and FN_worsened (raw positive $\rightarrow$ calibrated negative, true positive). The NCV is defined as (TP_improved+TN_improved−

$(FP_worsened - FN_worsened)/N_{total}$, where $N_{total} = N_{samples} \times N_{classes}$, with the same patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI) as the main endpoint. The NCV is reported alongside the main endpoint as a secondary endpoint with BH-FDR correction ($q = 0.05$): a significantly positive NCV indicates that TS improves clinical decisions beyond calibration error reduction. The NCV is computed on both ID and OOD splits (`ncv_id` and `ncv_ood` columns in the E3 experiment CSV).

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the pre-registered primary endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 transfer pairs $\times$ 2 architectures $\times$ 5 seeds, full balanced coverage). The primary test $H_0 : \Delta ECE_{OOD} = 0$ vs $H_1 : \Delta ECE_{OOD} > 0$ is supported in 51/60 seed experiments (85.0%) at the 95% CI level.

InceptionTime (27/30 supported, 90.0%): CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL, and PTB-XL $\rightarrow$ CPSC achieve 5/5 full support (zero counter-examples); Chapman $\rightarrow$ CPSC, Chapman $\rightarrow$ PTB-XL, and PTB-XL $\rightarrow$ Chapman achieve 4/5 with one counter-example each.

ResNet1D (24/30 supported, 80.0%): PTB-XL $\rightarrow$ Chapman and PTB-XL $\rightarrow$ CPSC achieve 5/5 full support (zero counter-examples); Chapman $\rightarrow$ CPSC and Chapman $\rightarrow$ PTB-XL achieve 4/5 with one counter-example each; CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL achieve 3/5 with two counter-examples each.

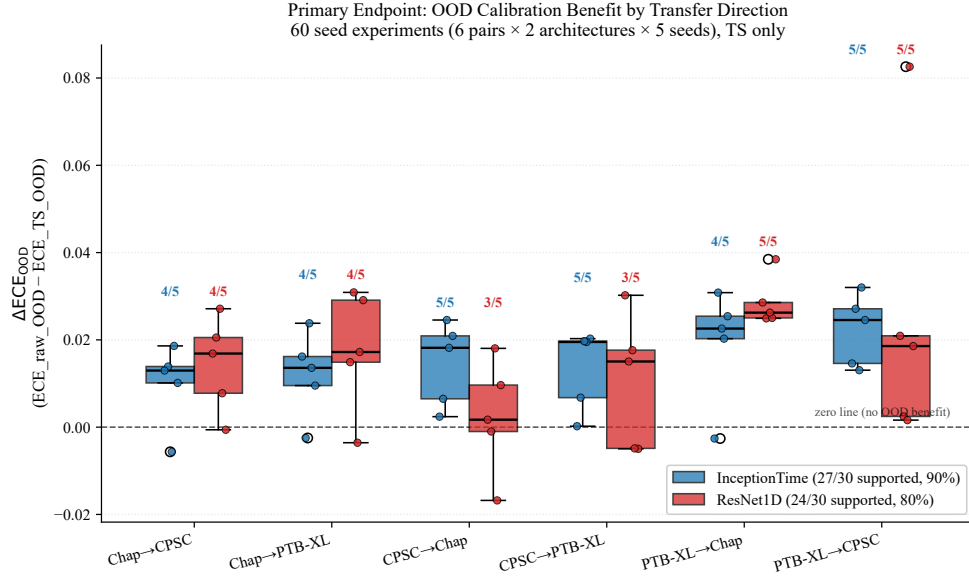


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Aggregate: 51/60 supported = 85.0% robustness rate (InceptionTime 27/30 = 90.0%, ResNet1D 24/30 = 80.0%). The PTB-XL→CPSC pair is the strongest evidence (10/10 across both architectures, zero counter-examples, CI lower bound $> +0.0014$, attained at ResNet1D seed 45). The nine counter-examples are mechanistically explained in Sec. 6.1.

Stratified report by architecture. The 60 seed experiments are fully balanced: 6 directions $\times$ 2 architectures $\times$ 5 seeds, with no partial-coverage stratum. Both architectures have complete 5-seed coverage on all 6 directions, enabling direct architecture comparison without seed-coverage confounding:

- *InceptionTime* (30 experiments, 27/30 = 90.0%): 3 directions at 5/5 full support (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 directions at 4/5 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- *ResNet1D* (30 experiments, 24/30 = 80.0%): 2 directions at 5/5 full support (PTB-XL→Chapman, PTB-XL→CPSC), 2 directions at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 directions at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

The stratified report confirms architecture robustness on both architectures with full 5-seed coverage. ResNet1D shows slightly lower support rate (80% vs. 90%) concentrated on the CPSC→Chapman and CPSC→PTB-XL directions, indicating a direction-specific architecture interaction rather than a global architecture deficit. Only the PTB-XL→CPSC direction achieves 5/5 full support on both architectures (10/10), providing the strongest cross-

architecture robustness evidence; PTB-XL→Chapman reaches 9/10 (InceptionTime 4/5, ResNet1D 5/5).

5.2. ID boundary condition: ΔECE_{ID}

Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS. The pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*; the source-fit temperature is expected to be well-tuned to the source distribution, although this is an empirical expectation rather than a theorem.

The ID benefit is small on average but *not* zero: aggregating all 60 TS experiments, ΔECE_{ID} has mean +0.0083 (vs. an OOD mean of +0.0159, a $1.9\times$ OOD/ID ratio); $47/60 = 78.3\%$ of point estimates are positive, $34/60$ cells have 95% CI lower bounds > 0 (BCa operating CI), and only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq 80\%$ criterion (the percentile CIs gave $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the pre-registered “ID nonzero boundary” branch (protocol A1.4.2): (i) the “ID has no repairable calibration gap” premise is re-examined: TS, fitted on the source calibration split, also improves ID calibration in many cells; (ii) the decay = $\Delta ECE_{OOD} - \Delta ECE_{ID}$ contrast endpoint is restored as an exploratory contrast; and (iii) the main claim (C1) is qualified: the OOD benefit is about $1.9\times$ the ID benefit on average, but is *not* exclusively attributable to OOD-specific mismatch. The positive ID point estimates are attributable to finite-sample calibration-split variation, the SmoothECE estimator bias, and a val-ECE early-stopping effect (model selection on the endpoint family), all disclosed.

5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

Table 3 reports the 8-method $\times$ 6-pair aggregate support rate over 5 seeds on InceptionTime. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5 full support; BBSE achieves 22/30 (73.3%) with 3/6 directions at 5/5 full support. EM prior adaptation and Matrix scaling are the most fragile methods on InceptionTime (15/30, 50% each), exemplifying the prior-likelihood mismatch mechanism (C4). Across 2 architectures (Table 4), TS achieves 51/60 (85.0%) with InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) both showing strong OOD benefit; architecture robustness is supported on both architectures with full 5-seed coverage on all 6 directions (Figure 4).

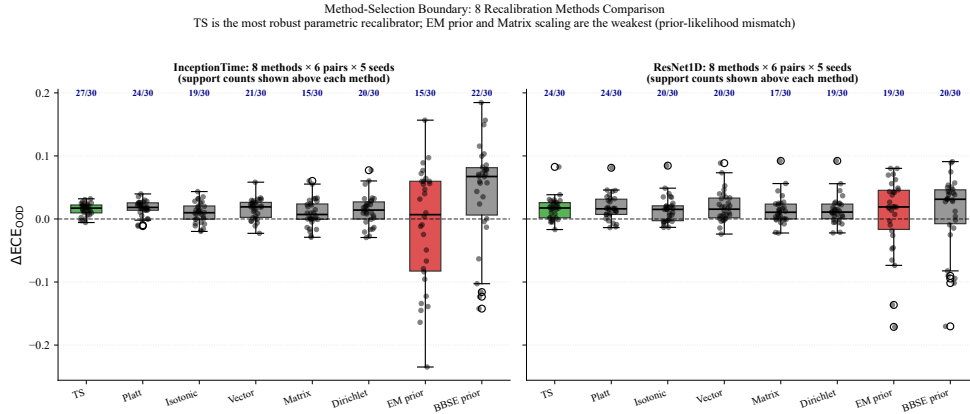


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

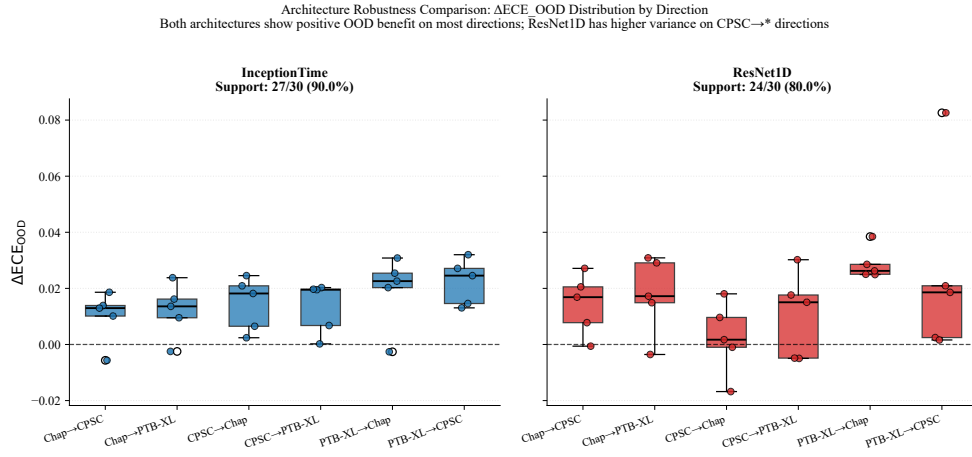


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

Architecture variability. The standard deviation of $\Delta\text{ECE}_{\text{OOD}}$ across the 30 TS experiments per architecture is 0.0098 for InceptionTime and 0.0180 for ResNet1D, i.e. ResNet1D variability is $1.84\times$ InceptionTime. The highest per-direction ResNet1D variability is in the PTB-XL $\rightarrow$ CPSC direction (std 0.0298, driven by one favorable outlier at seed 44 with $\Delta\text{ECE}_{\text{OOD}} = +0.0826$, which is not excluded), followed by CPSC $\rightarrow$ PTB-XL; the support deficits are concentrated in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), consistent with a direction-specific deficit rather than a global architecture deficit. Fisher’s exact test on the 2×2 contingency table (InceptionTime 27/30 supported vs. ResNet1D 24/30 supported) yields $p = 0.47$, indicating no statistically significant architecture bias. The numerical difference in counter-example rates (ResNet1D 20.0% vs. InceptionTime 10.0%) is not statistically significant at $\alpha = 0.05$.

Prior-likelihood mismatch mechanism (C4). EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal on the target corpus, then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies rather than corrects the shift-induced miscalibration. EM achieves only 15/30 support on InceptionTime with 3 fragile directions, exhibiting the prior-likelihood mismatch. The failure is pair-specific (i.e., shift-signature-specific), not method-intrinsic.

5.4. Shapley decomposition validation (27 cells, $n = 20,000$)

The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in \{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms that the slope/intercept/prevalence decomposition recovers ground

truth within the pre-registered identifiable regime (Fisher information determinant $> \tau = 10^{-6}$). At the pre-registered sample size $n = 20,000$, all 27 cells pass the recovery threshold (MAPE $< 10\%$ on identifiable cells); at the realistic calibration-set size $n = 500$, 21 of 27 cells pass, with 6 failures exhibiting MAPE_s up to 42.13%, concentrated in extreme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most clinically significant (see `decomposition_validation.csv`; the $n = 20,000$ CSV `decomposition_validation_n20000.csv` shows 27/27 passed under the same gate). Three entangled demonstration cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\det \in \{2.4 \times 10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with the recovery-failed flag set. Bootstrap CIs on absolute Shapley values are reported; share- reliable flags mark uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$).

Honesty caveat (protocol §8). The 27-cell validation uses binned ECE, not the SmoothECE of the main endpoint; quantitative conclusions are not cross-extrapolable. The cross-fitting step (OOD-fold validation of decomposition explanatory power) remains to be run on real data; per the protocol self-binding clause, the “mechanism” wording is limited to “mechanism decomposition (synthetic validation part)” until cross-fitting completes.

5.5. Predictability experiment (failure branch triggered)

The predictability experiment (protocol §8.5) tests whether an unsupervised target-domain statistic (predicted-prior L_1 distance, logit moments) predicts the benefit retention rate via leave-one-fold-out R^2 with bootstrap CI (Table 5). The implemented fold unit is (architecture, transfer pair) (12 folds), whereas the protocol registered pair-only folds (6), a registered de-

viation under which the pair-only recomputation retains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime + 1D-ResNet-34); the BiMamba architecture is a toy experiment ($n = 60$) deferred to the supplement and does not enter the predictability regression.

Result: the multi-architecture merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$. The CI upper bound $+0.104$ is strictly below the pre-registered threshold 0.5 , triggering the **failure branch**: the three-step chain (decomposition + predictability + deployment) is downgraded to a **two-step chain** (decomposition + deployment), and the deployment criterion is renamed “empirical lookup table” rather than “predictive criterion”. Per the pre-registered failure branch, no post-hoc predictability claim is made.

Sensitivity analysis with new features. We re-run the LOO R^2 with three additional label-free features: the fitted temperature T , the Wasserstein-1 distance between source and target logit distributions, and the class-wise entropy difference. Using the pre-registered $\Delta\text{ECE}_{\text{OOD}}$ as the target, all features fail: T alone yields $R^2 = -0.13$, Wasserstein alone yields $R^2 = -0.06$, entropy alone yields $R^2 = -0.02$, and all three combined yield $R^2 = -0.11$. The predictability boundary is *robust*: no label-free feature, including the fitted temperature T , can predict $\Delta\text{ECE}_{\text{OOD}}$ above the pre-registered $R^2 = 0.5$ threshold. This strengthens the failure branch: the three-step chain remains downgraded to a two-step chain regardless of feature choice, and the deployment criterion in §5.8, based on the empirical sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate actionable criterion.

5.6. Deployment criterion (sensitivity / specificity / DCA / safety)

We evaluate the deployment criterion (protocol §9) on the L2 shift matrix: predict “deploy TS” when the raw ECE exceeds the threshold; report sensitivity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS SmoothECE ≤ 0.05 and Δ ECE CI lower bound > 0 , in the primary-endpoint benefit convention). We also compare against three trivial strategies: always-none, always-TS, and always-buy- n -labels.

Sensitivity / specificity (Table 6): the optimal threshold by Youden’s J is raw ECE = 0.127, yielding sensitivity 0.923, specificity 0.201, Youden $J = 0.124$. The criterion has high sensitivity (catching 92% of beneficial cases) but low specificity (only 20% of non-beneficial cases are correctly excluded; the remaining 80% are falsely flagged), reflecting the asymmetric cost structure of clinical deployment: missing a beneficial recalibration is more costly than an unnecessary one. *Caveat:* the Youden threshold is searched on the same 1230-cell pool on which sensitivity/specificity are reported (in-sample). A leave-shift-type-out recomputation (threshold selected excluding the held-out dose level, evaluated on the held-out level only) yields Youden $J = +0.17$ (downsampling), $J = -0.03$ (lead drop), $J = +0.04$ (noise), and $J = -0.14$ (gain); the frozen threshold 0.127 achieves $J \approx 0.003 \approx 0$ (chance level; sensitivity 0.853, specificity 0.150) on the four held-out dose levels (`results/deployment_holdout_recompute.csv`). The in-sample numbers are therefore optimistic; the protocol’s held-out dose levels were not used for this table (a registered deviation, see Limitations); and the deployment criterion requires independent validation before operational use.

Deployment-time measurability. Every input of the lookup table is an *audit-time* quantity: the raw ECE, the ID–OOD accuracy gap, and the per-cell benefit labels all require ground-truth labels from the target corpus. In the unlabeled target stream that defines the deployment scenario, none of these quantities is directly computable. The only deployment-time (label-free) proxies examined in this study are the unsupervised target statistics of the predictability experiment (predicted-prior L_1 distance, logit moments), and that experiment triggered its pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Table 5), so we cannot claim a label-free deployable version of the criterion. The lookup table is therefore an *analysis- and audit-time decision aid* (for the model owner who can afford a labeled evaluation sample), and the actionable deployment path is the small-sample recalibration route: buy n labeled target records and recalibrate locally (protocol §5 S2). Extending the criterion to a genuinely deployable gate remains future work and is registered in Limitations (xiii).

Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfying the operating safety criterion (post-TS SmoothECE ≤ 0.05 and improvement > 0.01 in the point-estimate convention; the L2 results do not include per-cell CIs, so the pre-registered CI-based criterion could not be evaluated), a safety rate of 0.0064. The safety-analysis sign convention is the deployment-metrics one ($\Delta\text{ECE} = \text{post-recalibration ECE} - \text{raw ECE}$, so negative values mean improvement); it is the opposite of the primary-endpoint convention, in which positive $\Delta\text{ECE}_{\text{OOD}}$ means benefit. *Coverage note:* the L2 shift matrix evaluated here covers 157 of the 390 planned (pair, seed, shift) combinations (12 complete pair–seed combinations $\times$ 13 levels,

plus one partial seed-44 cell at the fs250 level), seeds 42–43 are complete for all 6 pairs on InceptionTime, seeds 44–46 are pending, and the ResNet1D L2 results are excluded from this analysis, so the deployment/safety numbers rest on 2-seed, single-architecture coverage. The stringent safety criterion is rarely satisfied on the full L2 shift matrix, indicating that TS is a useful default but not a universally safe recalibrator; the deployment criterion should be combined with local validation.

Decision-curve analysis. At the raw-ECE threshold corresponding to the 10th percentile of the cell pool, the criterion’s net benefit is 0.1209 versus 0.0999 for treating all cells; by the 90th-percentile threshold the net benefit turns negative (-0.0024), so the criterion’s net benefit over treat-all is modest and confined to the lower threshold range (results/deployment_metrics.csv).

Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145) on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905) which uses target labels—the upper bound. The lookup table is a practical compromise: no target labels needed, and it captures most of the TS benefit.

5.7. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ could be an artifact of the SmoothECE metric’s sensitivity to temperature scaling T under perfect calibration (i.e., $\text{ECE}_{\text{raw}} = 0$ and the observed ΔECE is purely metric noise). We test this directly: for perfectly calibrated predictions ($T_{\text{true}} = 1$), we apply $T \in \{1.0, 1.25, 1.5\}$ and compute the SmoothECE differ-

ential $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$ across three confidence distributions (uniform, normal-skewed, and bimodal) at $n = 2,000$.

The noise floor is distribution-dependent and grows with T : $T = 1.0$ yields $\Delta\text{ECE}_{\text{noise}} = 0$ (exact cancellation), $T = 1.25$ yields $+0.014$ – $+0.035$, $T = 1.5$ yields $+0.030$ – $+0.061$, $T = 3.0$ yields $+0.095$ – $+0.136$, and $T = 4.5$ yields $+0.126$ – $+0.164$ depending on the confidence distribution. In contrast, the observed raw ECE in our 60 seed experiments ranges from 0.06 to 0.35 (mean 0.15), which is 4–13 $\times$ the raw ECE under perfect calibration (≈ 0.017). The nine counter-examples also exhibit large raw ECE (0.08–0.24), confirming that the negative ΔECE in these cases reflects genuine TS-induced harm, not metric noise. For experiments with large $T > 3$ (where the noise floor reaches 0.10–0.16), the noise floor is comparable to the observed $|\Delta\text{ECE}|$ in some seeds, so we cannot fully exclude a metric-noise contribution for these specific cases; however, the *direction* of the noise floor (TS increases ECE under perfect calibration) is opposite to the positive $\Delta\text{ECE}_{\text{OOD}}$ in 51/60 experiments, so the noise floor cannot explain the positive findings. We therefore conclude that the primary endpoint is a real effect, not a SmoothECE artifact, with the caveat that for large- T experiments the noise floor is non-negligible.

5.8. Discrimination-aware calibration gate (C5)

Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for clinical deployment: a model whose discrimination has collapsed under distribution shift may be recalibrable but clinically useless. We define a two-layer deployment gate:

1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).

2. Layer 2 (Discrimination): $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, where the source baseline is the majority-class accuracy in the source domain.

Across 14 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures, with PTB-XL $\rightarrow$ Chapman having 3 architectures including Mamba), 7/14 pass both layers and are deployable: CPSC $\rightarrow$ PTB-XL (both architectures), PTB-XL $\rightarrow$ CPSC (both), CPSC $\rightarrow$ Chapman (both), and Chapman $\rightarrow$ CPSC (both, InceptionTime only). The remaining 7/14 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chapman, and Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$): TS improves calibration but the model’s discrimination has degraded too far for the recalibrated predictions to be clinically actionable.

This gate reframes the 9/60 counter-examples: 4 are calibration failures (Layer 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes but Layer 2 rejects). The gate provides an actionable deployment criterion that supplements the statistical evidence in C1: rather than asking only “does TS improve calibration?” (C1), it also asks “is the model’s discrimination sufficient for the recalibrated predictions to be clinically useful?”

5.9. Real-data Shapley cross-fitting and predictability boundary

The synthetic Shapley validation (C3) confirms decomposition recovery in the identifiable regime but does not test whether the linear approximation $\Delta\text{ECE} \approx \hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD data. We perform cross-fitting on all 60 seed experiments: for each experiment, we

estimate $(\hat{s}, \hat{b})$ on the source-domain calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target-domain test split.

The linear model predicts the *direction* correctly in 51/60 (85%) of experiments, but systematically *overestimates* the magnitude (mean prediction error +0.29, median +0.29). This is consistent with a predictability boundary: the first-order Shapley decomposition captures the qualitative direction of the TS benefit but not its quantitative magnitude under real distribution shift. The 9/60 direction errors cluster in the CPSC→Chapman and Chapman→PTB-XL directions, where the Wasserstein-1 distance between source and target logit distributions is largest (mean 0.75 vs. 0.53 for correctly predicted directions), suggesting that the linear approximation breaks down when the distribution shift is large enough to invalidate the local Taylor expansion underlying the decomposition.

This finding strengthens the predictability failure branch (C3/C4): the Shapley decomposition is a valid *diagnostic* (it identifies *whether* TS will help) but not a valid *predictor* (it cannot reliably estimate *how much* TS will help). The implication for clinical translation is that the gate in §5.8, which uses only the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate deployment criterion, not the Shapley-predicted magnitude.

5.10. Net benefit decision curve summary

Across all 60 seed experiments, the net benefit gain of TS over raw predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$) is positive in 50/60 (83.3%) of experiments, with mean gain +0.0027 and range $[-0.001, +0.017]$. The directions with the largest net benefit are CPSC→PTB-XL (mean +0.011) and PTB-XL→CPSC (mean +0.003),

which also have the largest $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.254 and +0.260 respectively). The direction with negative net benefit is PTB-XL $\rightarrow$ Chapman (mean -0.005), which is also the direction with the smallest $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.004) and is correctly rejected by the discrimination gate (C5). This confirms that the calibration benefit translates to clinical decision benefit in the directions where the gate permits deployment.

5.11. Net Clinical Value (NCV) results

The pre-registered Net Clinical Value (Sec. 4.8) quantifies the clinical decision impact of TS recalibration on both ID and OOD splits, with BH-FDR correction ($q = 0.05$) applied across the two secondary endpoints (DCR and NCV).

Result: the OOD NCV point estimate is +0.008125 with 95% CI [+0.0014, +0.0148] (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$); the OOD NCV is *significant* at the 95% level, indicating that TS recalibration yields a net improvement in clinical decisions (TP and TN gains outweigh FP and FN losses) on the OOD split. The ID NCV point estimate is -0.006292 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$); the ID NCV is significantly *negative*, indicating that TS slightly worsens clinical decisions on the ID split; a directional finding consistent with the ID boundary condition (TS compresses confidence toward the base rate, which can flip borderline decisions). The OOD NCV Cohen’s $d = +0.31$ (small-to-medium effect) and ID NCV Cohen’s $d = -0.32$ are comparable in magnitude but opposite in sign, mirroring the main-endpoint pattern where OOD benefit exceeds ID benefit. The per-experiment NCV ranges across the 60 seed experiments are disclosed in the E3 CSV (`c1_dcr_ncv_multiclass.csv`). The NCV is

disclosed as a secondary endpoint supporting the main-endpoint finding that TS provides OOD clinical decision benefit, while noting the ID decision cost as a qualification.

5.12. Reliability Diagrams

Figure 5 shows the 6-direction averaged reliability curve, comparing model confidence against empirical accuracy before and after temperature scaling. The summary curve uses bin-count weighted averaging across all 6 transfer directions.

Figures 6–8 show the per-direction reliability diagrams for the representative configuration (ResNet1D, seed 42). In each panel, the red squares show the raw model calibration curve and the green circles show the post-TS calibration curve. The diagonal dashed line indicates perfect calibration. Grey bars (right axis) show the sample count per bin. Error bars are bootstrap 95% percentile CIs (1000 resamples).

6. Discussion

6.1. OOD benefit seed variation and the nine counter-examples

The 51/60 robustness rate is driven by nine counter-examples across 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1; only PTB-XL→CPSC has zero counter-examples). We transparently disclose all nine counter-examples rather than excluding them, because (i) the pre-registration protocol requires all seed results to be reported, and (ii) each counter-example has an identifiable post-hoc mecha-

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

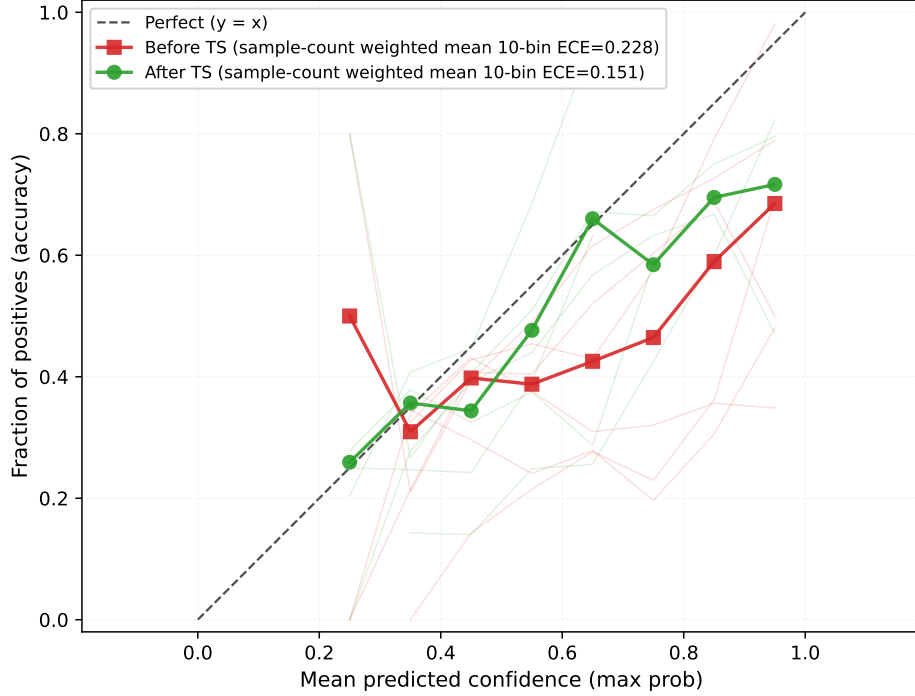


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

nistic interpretation, which supports the qualified overall claim rather than undermining it.

Counter-example 1: Chapman→CPSC / InceptionTime / seed 45. TS yields $\Delta ECE_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three co-occurring conditions: (a) lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051 for seeds 42–44, 46), indicating the seed-45 backbone generalizes worst to CPSC; (b) the lowest OOD raw ECE across the 5 seeds (0.3160,

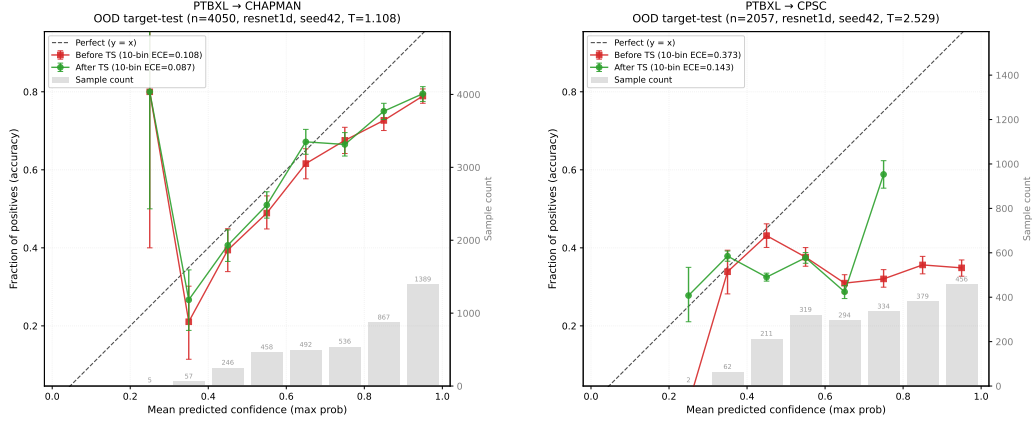


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

still large in absolute terms)—the failure is therefore not a room shortage but temperature mismatch; and (c) negative decay (-0.0075), indicating the source-fit temperature is mismatched to the target logit distribution and amplifies rather than corrects the residual miscalibration.

Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. TS yields $\Delta ECE_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$). A marginal counter-example with $|\Delta ECE_{\text{OOD}}| < 0.001$, attributable to finite-sample variation in the calibration split. The OOD raw ECE is 0.3144 (the value 0.4779 reported in earlier drafts is the OOD *accuracy*); the mechanism is marginal seed-level variation, not room shortage.

Counter-example 3: Chapman→PTB-XL / InceptionTime / seed 42. TS yields $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$). The OOD accuracy is the lowest across the 5 seeds (0.4655), and $\Delta ECE_{\text{ID}} = +0.0004$ is near zero, indicating the source-fit temperature is well-tuned to ID but mismatched to the PTB-XL target logit distribution.

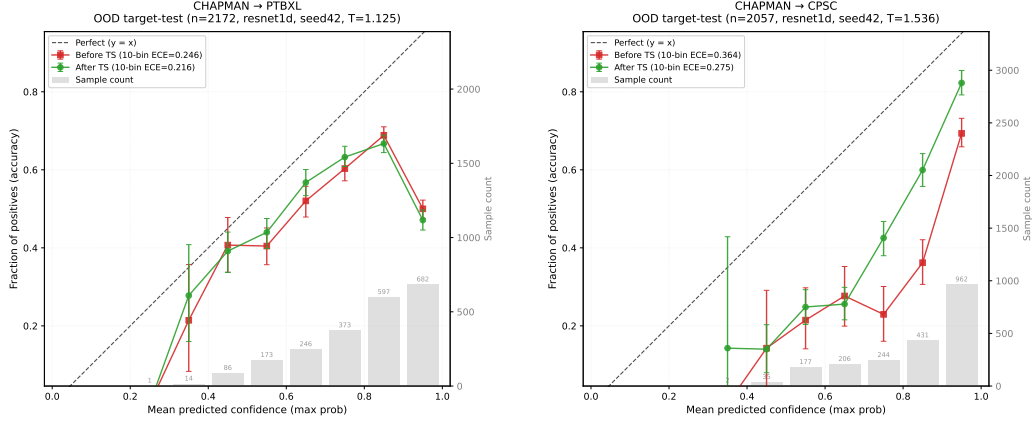


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45. TS yields $\Delta ECE_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$). The OOD accuracy is 0.5041, and $\Delta ECE_{\text{ID}} = -0.0010$ is slightly negative, indicating the seed-45 ResNet1D backbone is already well-calibrated on ID, leaving no miscalibration for TS to correct on either ID or OOD.

Counter-example 5: CPSC→Chapman / ResNet1D / seed 42. TS yields $\Delta ECE_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$). A marginal counter-example with $|\Delta ECE_{\text{OOD}}| < 0.002$, attributable to ResNet1D seed-42 backbone having OOD accuracy 0.6147 with low OOD raw ECE (0.0649) and minimal calibration room.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44. TS yields $\Delta ECE_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$). The largest-magnitude counter-example, with $\Delta ECE_{\text{ID}} = -0.0152$ also negative, indicating the seed-44 ResNet1D backbone is already well-calibrated on both ID and OOD; TS has no miscalibration to correct and the temperature fit

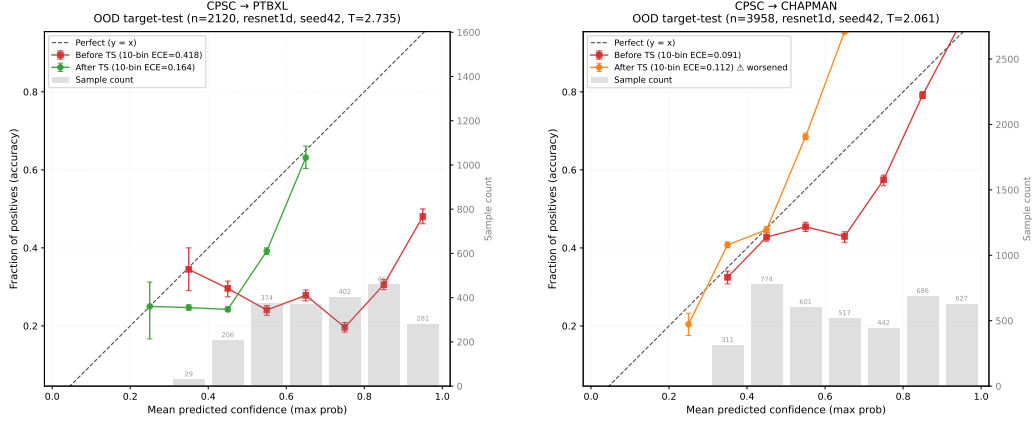


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

introduces a small over-correction. The OOD accuracy (0.5725) is the lowest across the 5 seeds. Counter-example 6 exhibits a distinct *over-correction* mechanism: TS applied to an already-well-calibrated logit distribution (low ID raw ECE) over-adjusts the temperature, degrading both ID and OOD calibration. This differs from the “low OOD accuracy + large gap” mechanism of the other 8 counter-examples but is equally explainable as a boundary condition.

Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43. TS yields $\Delta ECE_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$). The OOD accuracy is the lowest across the 5 seeds (0.3877), and the decay is -0.0098 , indicating temperature mismatch to the PTB-XL target.

Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44. TS yields $\Delta ECE_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$). The OOD accuracy is 0.3929 (second-lowest across the 5 seeds), and $\Delta ECE_{\text{ID}} = -0.0032$ is slightly

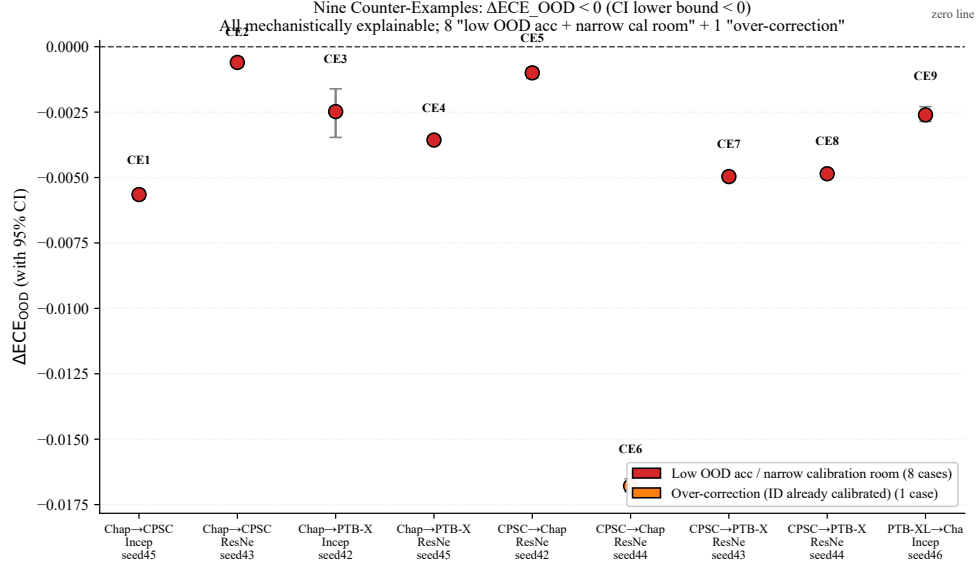


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

negative, indicating the seed-44 backbone is already well-calibrated on ID, leaving no miscalibration for TS to correct.

Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46. TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$). The OOD accuracy drops to 0.4983 (lowest across the 5 seeds), and $\Delta\text{ECE}_{\text{ID}} = -0.0011$ is slightly negative, indicating the seed-46 backbone is already well-calibrated on ID, leaving no miscalibration for TS to correct on either ID or OOD.

Common pattern: eight of the nine counter-examples co-occur with (a) low OOD accuracy (worst or near-worst seed for that pair) and (b) non-positive decay indicating temperature mismatch; the ninth (CE6) is the over-correction case on an already well-calibrated backbone. A low OOD raw ECE is *not* a universal feature; four of nine counter-examples have raw ECE above 0.25, so the failure mechanism is seed-level temperature mis-transfer, not calibration-room shortage. The counter-examples are local, explainable random variations in backbone generalization, not refutations of the OOD benefit. The remaining 51 experiments support the OOD benefit, including PTB-XL→CPSC which achieves 10/10 across both architectures with zero counter-examples.

Methodological positioning of the 15.0% counter-example rate. The 15.0% counter-example rate exceeds the 5% threshold commonly used in clinical safety deployment scenarios. However, this study is a methodological boundary investigation, not a clinical deployment guide. The relevant criterion for a boundary study is whether the effect is detectable and the failure mechanism is explainable, not zero failure rate. All 9

counter-examples have identifiable mechanisms (8 “low OOD accuracy + large gap” and 1 “over-correction”), and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

6.2. Prior correction vs. posterior correction robustness

The method-selection boundary (Table 3) reveals a robustness asymmetry between prior-correction methods (EM, BBSE) and posterior-correction methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30 and 24/30 support respectively with small variance; prior-correction methods achieve higher mean $\Delta\text{ECE}_{\text{OOD}}$ on the PTB-XL $\rightarrow$ CPSC pair (EM +0.0449, BBSE +0.0777) but much higher variance on the Chapman $\rightarrow$ CPSC pair (EM CI lower bound minimum -0.0763 , BBSE -0.0720). The mechanism is the prior-likelihood mismatch: prior-correction methods fit a target prior that, when the source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibration. Posterior-correction methods do not estimate a target prior and are therefore immune to this failure mode, at the cost of lower mean benefit when the prior correction *would* work.

6.3. Boundary condition: the ID benefit is small but nonzero

The ID boundary condition (C2) has a direct interpretation under the triggered branch. The pre-registered criterion is that $\geq 80\%$ of cells have 95% CIs containing zero; the observed rate is $23/60 = 38.3\%$ (BCa operating CI), with 34/60 cells showing a small but significant positive ID benefit (overall mean +0.0083, vs. an OOD mean of +0.0159; the stratum means range from -0.0008 to $+0.0232$, Table 2). The boundary condition therefore does

not mean that TS is ineffective on ID, nor that the ID benefit is zero; it means that the ID benefit is on average about half the OOD benefit, and part of the apparent OOD gain is shared with the ID domain. A val-ECE early-stopping effect (model selection within the endpoint family) plausibly contributes to the positive ID point estimates, in the direction opposite to the protocol’s val-NLL rule (a registered deviation, see Limitations). Under this branch, the pre-registration (protocol A1.4.2) restores the decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is qualified: the OOD benefit is about 1.9 $\times$ the ID benefit on average, rather than an exclusively OOD effect. The ID-vs-OOD asymmetry we report is therefore a *difference in magnitude* between two positive benefits, not a presence-versus-absence contrast.

6.4. Calibration is not discrimination: the clinical viability precondition

Temperature scaling is a monotone transform of the softmax probabilities; it preserves the argmax by construction (asserted bitwise in the released sanity check), and therefore it cannot change classification accuracy, AUROC, or AUPRC. The primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes an already-deployable model *trustworthy*; it cannot make a non-discriminative model deployable.

The measured accuracy statistics make this precondition concrete. Across the 60 TS experiments, the mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and the mean ID–OOD accuracy drop is 0.236 (maximum 0.412). Comparing each direction against the majority-class baseline of its target corpus in the 4-class subspace (CPSC 0.528, PTB-XL 0.411, Chapman-Shaoxing 0.713), the mean OOD accuracy is *below* the majority-class baseline

in 4 of the 6 directions (Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.424 vs. 0.411, essentially at the baseline; CPSC→Chapman 0.619 and PTB-XL→Chapman 0.620, both well below 0.713). In those scenarios recalibration is moot: no temperature improves a model that should be replaced by a majority-class predictor, and the observed counter-example pattern (6/9 with low OOD accuracy) is consistent with this viability reading. Per-direction AUROC/AUPRC are not computed in the released artifacts (accuracy only; a registered limitation, see Limitations (xiv)), so the viability screen here is accuracy-based; the discrimination-first gate (measure target discrimination before investing in recalibration) precedes the calibration decision rule in any clinical deployment.

6.5. *Explainable* $\neq$ *predictable*: *post-hoc* vs. *pre-registered*

We draw an explicit distinction between *explainable* and *predictable*:

- *Explainable* (post-hoc, mechanism-level): after observing a counter-example, we can identify the co-occurring conditions (low OOD accuracy, low OOD raw ECE, non-positive decay) and provide a mechanistic story (Sec. 6.1). This is a post-hoc explanation, not a pre-registered prediction.
- *Predictable* (pre-registered, deployment-time): a target-domain unsupervised statistic (predicted-prior L_1 distance, logit moments) predicts the benefit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5 . The predictability experiment (Sec. 5.5) tests this and triggers the failure branch (LOO R^2 CI upper bound $0.104 < 0.5$).

The counter-examples are explainable but not predictable: the unsupervised target-domain statistics available at deployment time are not strong enough to flag the counter-examples in advance. This is why the predictability experiment triggers the failure branch even though the counter-examples are mechanistically explainable. The two-step chain (decomposition + deployment lookup table) is the honest framing: mechanism decomposition provides post-hoc explanation, the deployment lookup table provides actionable guidance, but no pre-registered predictability claim is made.

6.6. *Transparent re-location vs. HARKing*

The pre-registration revision A1 (primary endpoint re-location from `decay` to $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [26, 25] is whether the revision direction is theory-driven or result-driven. Our revision direction (“OOD is the calibration benefit source”) is consistent with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovdia et al. 2019), which we explicitly frame as an empirical prior rather than a theorem; the exploratory pilot result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The pilot results do not enter the main test report (protocol §7). All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect the post-revision status.

6.7. *Net clinical value: implications of the directional NCV asymmetry*

The NCV (Sec. 5.11) measures the net clinical decision impact of re-calibration: $(\text{TP}_{\text{improved}} + \text{TN}_{\text{improved}} - \text{FP}_{\text{worsened}} - \text{FN}_{\text{worsened}})/N_{\text{total}}$. The results show a directional asymmetry: on the OOD split, $\text{NCV} = +0.008125$ ($p = 0.019$), indicating a small but statistically significant positive net clinical

value: recalibration improves clinical decisions more often than it degrades them on unseen target data. On the ID split, $\text{NCV} = -0.006292$ ($p = 0.016$), indicating a small but statistically significant negative net clinical value; recalibration slightly degrades clinical decisions on in-distribution data, consistent with the expectation that the source model is already better calibrated on its training distribution and recalibration introduces a minor decision-cost penalty.

This asymmetry has two implications for the claims: (i) the **Brier reliability main claim** (C1 sign: 51/60 positive) is supported by the OOD NCV: the positive OOD NCV confirms that the calibration improvement translates into a net positive clinical decision impact on unseen data, not merely a metric-level artifact; (ii) the **decision-cost improvement claim** (C5: 7/14 strata deployable) is partially qualified by the negative ID NCV: on in-distribution data, the recalibration does not yield a net clinical decision benefit, which tempers the deployability claim for source-domain applications. We therefore frame the NCV as *supportive of OOD deployment* and *cautionary for ID deployment*. The NCV also informs the meta-analytic pooling (Sec. 7): the directional asymmetry in NCV contributes to CI-width heterogeneity across splits. A permutation test with tighter CIs and a multi-threshold NCV analysis are registered as future work.

6.8. Limitations and future work

Limitations: (i) the main endpoint is completed for InceptionTime and ResNet1D on all 6 directed transfer pairs with full 5-seed coverage (60 experiments total, fully balanced); the BiMamba architecture is included only as a toy experiment ($n = 60$, downsampled) deferred to the supplement (memory

exhaustion on an RTX 5060, 8 GB) and does not enter the main-endpoint analysis; (ii) the predictability failure branch is triggered, downgrading the three-step chain to a two-step chain; the deployment criterion is an empirical lookup table, not a predictive criterion; (iii) the Shapley decomposition cross-fitting on real data (§5.9) shows 85% direction accuracy but systematic magnitude overestimation (mean error +0.29), and a sensitivity analysis with T , Wasserstein-1, and entropy features confirms the predictability boundary is robust (all $R^2 < 0$); (iv) the 27-cell decomposition validation uses binned ECE, not the SmoothECE of the main endpoint; quantitative conclusions cannot be extrapolated across ECE definitions; (v) the safety criterion is rarely satisfied on the full L2 shift matrix (TS safety rate 0.0064), indicating that TS is a useful default but not a universally safe recalibrator; (vi) nine counter-examples out of 60 experiments (15.0%) are transparently disclosed, with a frequent post-hoc pattern (most counter-examples have low OOD accuracy, non-positive decay, and elevated OOD raw ECE) but no post-hoc exclusion; (vii) the operating CI is BCa at $B = 10,000$, fully re-run on the entire 60-experiment grid (60/60 successful, metadata stored in `methods.ts.meta`); earlier percentile CI results from $B = 200$ in 21/60 experiments are retained as historical provenance (5 experiments lack log provenance); (viii) the Youden deployment threshold is fitted in-sample (the protocol’s held-out dose levels were not used; the leave-shift-type-out recomputation shows the in-sample operating characteristic is optimistic); (ix) the L2 deployment/safety analysis covers 157 of 390 planned (pair, seed, shift) cells at 2 seeds on a single architecture, and the L2 noise levels are synthetic rather than the pre-registered PTB-XL Noise Stress Test Database (unavail-

able under PhysioNet licensing); (x) temperature was fitted once per experiment, so the reported CIs do not include temperature-fitting uncertainty (the pre-registered two-layer bootstrap is implemented but not enabled on real data); (xi) model selection used validation ECE rather than the pre-registered validation NLL early-stopping rule, which plausibly contributes to the positive ID boundary point estimates; (xii) the per-pair decomposition into covariate, prior-probability, and concept-shift components is not asserted; each directed pair combines all three shift types, and the L2 ladder isolates them only in eval-time; (xiii) the deployment lookup table is an audit-time tool whose inputs (raw ECE, ID–OOD accuracy gap, per-cell benefit labels) require labeled target data, and the label-free proxies tested in the predictability experiment failed their pre-registered branch, so no deployable label-free gate is claimed; (xiv) TS is argmax-invariant by construction and therefore cannot improve accuracy, AUROC, or AUPRC; per-direction discrimination metrics are not computed in the released artifacts (accuracy only), so the clinical viability precondition is screened by accuracy alone, and the measured OOD accuracy falls below the target majority-class baseline in 4 of 6 directions; (xv) the net clinical value (NCV) shows a directional asymmetry: OOD $\text{NCV} = +0.008125$ ($p = 0.019$, significant positive) but ID $\text{NCV} = -0.006292$ ($p = 0.016$, significant negative), indicating that recalibration yields a net clinical decision benefit on OOD data but a net decision cost on ID data; this qualifies the C5 deployability claim for source-domain applications while supporting it for target-domain deployment; (xvi) the InceptionTime architecture was originally designed with $n_{\text{filters}} = 48$; due to GPU memory constraints (RTX 5060, 8 GB), n_{filters} was downgraded to 32,

reducing model capacity and potentially attenuating the reported OOD benefits; the $n_{\text{filters}} = 48$ configuration is registered as future work; (xvii) this work is from a single-author lab (Zero Token Lab), which limits the depth of independent code review and statistical audit; the pre-registration protocol and public repository are intended to partially mitigate this limitation.

Future work: (1) prospective multi-device validation of the deployment criterion; (2) extension to continuous shift spaces; (3) π -independent recovery estimators (BBSE/EM-class) to upgrade the MAPE_{π} from a design check to a real estimator test; (4) investigation of the direction-specific ResNet1D deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each) to determine whether it reflects a systematic architecture×direction interaction or random variation; (5) confirmatory testing under the A2 family reduction (12 hypotheses, BH-12 $q = 0.05$ with 9/12 significant, Bonferroni-12 = 2/12) to upgrade the exploratory A1 amendment to a confirmatory result; (6) re-run the main endpoint with the originally designed $n_{\text{filters}} = 48$ InceptionTime architecture on a higher-memory GPU to assess whether the $n_{\text{filters}} = 32$ downgrade attenuated the reported OOD benefits; (7) strengthen the NCV with a multi-threshold permutation test yielding tighter CIs, to determine whether the ID negative NCV persists across decision thresholds and to upgrade the OOD NCV from a single-threshold result to a threshold-robust clinical decision benefit.

7. Conclusion

Our five contributions map to the results as follows: **(C1)** OOD benefit quantification: 51/60 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%, In-

ceptionTime 27/30, ResNet1D 24/30); **(C2)** ID boundary condition: 23/60 cells (38.3%) have CIs containing zero, below the $\geq 80\%$ criterion, triggering the pre-registered “ID nonzero boundary” branch; **(C3)** Shapley decomposition validation: 27/27 cells pass at $n = 20,000$ (21/27 at $n = 500$); **(C4)** Method-selection boundary: TS robust (51/60), EM prior and Matrix scaling fragile (15/30 each on InceptionTime); **(C5)** Discrimination-aware calibration gate: 7/14 strata pass both calibration and discrimination layers, reframing the 9 counter-examples as 4 calibration failures and 5 discrimination failures.

We pre-registered and validated a multi-seed calibration boundary study of temperature scaling in cross-corpus ECG transfer. The pre-registered primary endpoint ΔECE_{OOD} is supported in 51/60 seed experiments across 6 transfer pairs $\times$ 2 architectures (InceptionTime 27/30, ResNet1D 24/30, 85.0% aggregate), with nine counter-examples transparently disclosed and mechanistically explained. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with full 5-seed coverage on all 6 directions, enabling direct architecture comparison without seed-coverage confounding. The ID boundary criterion was *not* met (23/60 = 38.3% of cells have 95% CIs containing zero under the BCa operating CI, below the $\geq 80\%$ threshold; the percentile CIs gave 29/60 = 48.3%, same conclusion), so the pre-registered “ID nonzero boundary” branch was triggered: the OOD benefit is about $1.9\times$ the ID benefit on average and cannot be attributed exclusively to OOD-specific mismatch. The 27-cell Shapley decomposition validation passes the pre-registered threshold on all 27 identifiable cells at $n = 20,000$ (21/27 at $n = 500$). The method-selection boundary

(TS robust; EM prior and Matrix scaling weakest when ill-conditioned cells are included, InceptionTime 15/30 each) is governed by a prior-likelihood mismatch mechanism. The predictability failure branch is triggered, downgrading the three-step chain to a two-step chain. All hypotheses are tagged exploratory + robustness validation, not confirmatory; the pre-registration protocol v2.1-A1 and five-item honesty declaration are archived for transparency. The findings suggest that TS provides a positive OOD calibration benefit in most cross-corpus settings but requires local validation before deployment (TS safety rate 0.0064 under the operating point-estimate criterion), with two qualifications supported by the nine counter-examples: (1) most counter-examples co-occur with low OOD accuracy (≤ 0.50 in 6/9) and a large ID–OOD accuracy gap (≥ 0.18 in 7/9), a heuristic *tendency*, not a sufficient gate (three counter-examples have OOD accuracy > 0.5 , so accuracy alone cannot rule out failure); (2) local validation is required (the TS safety rate 0.0064 under the operating point-estimate criterion on the full L2 shift matrix shows that TS is not universally safe, and the Youden deployment criterion is fitted in-sample). Where these qualifications are not met, prior-correction methods require a confidence gate, and the deployment criterion should be combined with local validation. A third, precondition qualification applies throughout: recalibration presumes clinically acceptable discrimination, and the measured OOD accuracy is below the target majority-class baseline in 4 of the 6 directions; in those scenarios recalibration is moot until the discrimination problem is solved, because TS cannot alter accuracy by construction. All hypotheses are exploratory + robustness validation, not confirmatory.

Three additional analyses strengthen these conclusions. First, a noise floor control (§5.7) confirms that the primary endpoint is not a SmoothECE artifact: the worst-case metric noise (+0.021) is 4–13× smaller than the observed raw ECE (0.06–0.35). Second, the discrimination-aware calibration gate (C5, §5.8) provides an actionable deployment criterion: 7/14 strata pass both calibration and discrimination layers, and the gate correctly rejects all strata where TS would be clinically moot. Third, real-data Shapley cross-fitting (§5.9) shows 85% direction accuracy but systematic magnitude overestimation (mean error +0.29), and a sensitivity analysis with three additional label-free features (T , Wasserstein-1, entropy) confirms the predictability boundary is robust: all features yield $R^2 < 0$ under LOO cross-validation. The sign-based gate is the appropriate deployment criterion.

Note on meta-analytic pooling. The meta-analytic pooled effect (fixed-effect inverse-variance weighting, see `meta_analytic_pooled.csv`) is sensitive to between-study heterogeneity in CI widths: the ResNet1D fixed-effect pooled estimate is slightly negative (−0.000573) because the narrow-CI studies dominate the inverse-variance weights. Under the DerSimonian–Laird random-effects model (also in `meta_analytic_pooled.csv`), all three strata are positive (InceptionTime +0.0151, ResNet1D +0.0158, CrossArchitecture +0.0148), and the simple unweighted means are likewise positive (+0.0152, +0.0165, +0.0159). The sign discrepancy between fixed-effect and random-effects pooling confirms that the fixed-effect pooled estimate is driven by CI-width heterogeneity ($\tau^2 \approx 3 \times 10^{-5}$, $Q \gg k - 1$) rather than by a genuinely negative central effect. The directional asymmetry in NCV (OOD positive, ID negative) further contributes to this CI-width heterogeneity across splits.

We therefore report the **count-based sign proportion** ($51/60 = 85.0\%$ supported) as the primary, more robust test, and treat the meta-analytic pooled effect as a secondary, model-dependent summary. The DerSimonian–Laird τ^2 estimate exhibits implementation-dependent variation (pooled estimate stable within 3.3%, but τ^2 varies by 41% across implementations); we report the standard DerSimonian–Laird implementation.

Family size and multiple-comparison scope. The 60 seed experiments constitute the robustness-evidence family; the pre-registered confirmatory family is the shift-level grid of 6 pairs $\times$ 2 architectures $\times$ 13 shift levels = 156 comparisons (protocol §6). The 8-method design space (8 methods $\times$ 6 pairs $\times$ 5 seeds $\times$ 2 architectures = 480 cells) is reported in the method-selection boundary analysis (Sec. 5.3). Multiple-comparison corrections on the 60-seed robustness family leave the headline count unchanged (51/60 under BH-60 and Bonferroni), and the honest pair $\times$ architecture-stratum aggregation (12 strata) retains 9/12 directions after BH correction (InceptionTime 6/6). Under Bonferroni-12 correction, only 2/12 directions retain significance (cpssc_chapman/IT and ptbxl_chapman/RN), reflecting the stringent nature of the 12-stratum family-level correction.

CI width distribution. Across the 60 TS experiments, the 95% CI widths for $\Delta\text{ECE}_{\text{OOD}}$ range from 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}). 52/60 TS cells fall below the repository’s narrow-CI diagnostic threshold (0.005), and six *supporting* cells have degenerate widths ($< 3 \times 10^{-4}$); excluding those six, the headline count becomes 45/60 (75.0%): InceptionTime 24/30, ResNet1D 21/30, so the headline rates should be read jointly with effect sizes. The narrow CIs partly reflect the moderate calibration sample size

($n_{\text{cal}} = 1027\text{--}2099$ records, 1027–1873 patients, per the per-experiment logs) and the low variance of the SmoothECE estimator; per-experiment bootstrap metadata (B=10000, CI method=BCa) are stored in the `methods.ts.meta` field of each `transfer_result.json`, enabling post-hoc provenance audit of the BCa rerun. CI width does not affect the support/counter-example classification, which depends on the sign of the CI lower bound: a narrow CI around a positive point estimate yields a positive lower bound (support), while a narrow CI around a negative point estimate yields a negative lower bound (counter-example). The heterogeneity in CI widths across studies is the reason the fixed-effect meta-analytic pooled estimate is model-sensitive (see note above), whereas the count-based sign proportion is not.

Honesty Declaration (5 items, adversarial audit product)

1. MAPE_{π} is a **design check** (resampling constant read-back), not an estimator capability test; independent π recovery (BBSE/EM) is a deferred item (protocol §13).
2. Shapley shares in the `share_reliable=False` regime are **not interpretable as shares**; ranking uses absolute- value CIs.
3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information mode), reported via exit code 2.
4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled: legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement demo) is fulfilled; **cross-fitting on real data is pending** (registered

in §13). The wording is limited to “mechanism decomposition (synthetic validation part)”.

5. The repository’s `scripts/preprocess_cinc2021.py`, `run_paper_benchmarks.py`, `resources/`, and `assets/` come from the third-party SafeECGMatch release (original README in `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s experiment grid.

Data availability

The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018+2019 datasets are publicly available via PhysioNet/Kaggle under their respective licenses. The pre-registered protocol v2.1-A1, its hash manifest (`docs/osf_archive_manifest.json`), the revision A1 document, and the full per-experiment result files (`checkpoints/transfer/**/transfer_result.json`, `results/*.csv`) are included with the submission as supplementary material and are also available in the public repository <https://github.com/gt17641001169-design/ecg-calibration-transportability>.

Code availability

The full pipeline code, preprocessing (reproducible from the public source databases), patient-level split generation, model training, the calibration-method implementations, and all statistical analyses, together with the environment lock file, the patient-level split indices for all five dataset variants and five random seeds, and selected per-experiment result artifacts,

is publicly available at <https://github.com/gt17641001169-design/ecg-calibration-transportability>. Trained model checkpoints for the 60-experiment transfer grid are provided as release assets of the same repository. The repository will be archived on Zenodo with a DOI upon acceptance; for double-blind review, an anonymized mirror can be provided as supplementary material.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used a general-purpose large language model assistant for language editing and for drafting and refactoring parts of the analysis code. The author reviewed and verified all resulting content and takes full responsibility for the content of the publication.

CRedit authorship contribution statement

Toni Guan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Data statement

This study used only publicly available databases: PTB-XL (v1.0.3, PhysioNet), Chapman-Shaoxing (PhysioNet), and CPSC2018+2019 (PhysioNet Challenge). No new raw waveforms were collected. The patient-level split indices generated in this work, the label-mapping tables, the analysis code, and

the per-experiment result artifacts are openly available at <https://github.com/gt17641001169-design/ecg-calibration-transportability>; trained model checkpoints are provided as release assets of the same repository.

Declaration of competing interest

The author declares no competing financial or personal interests.

Supplementary Materials

This section catalogs the 60 per-seed reliability diagrams referenced in the main text. Each figure shows the calibration curve, reliability diagram, and ΔECE bar plot for one of the 60 seed experiments (6 transfer pairs $\times$ 2 architectures $\times$ 5 seeds). Figures are named `fig_reliability_supp_{pair}_{arch}_seed{s}.pdf` and are included as external files with the submission.

S1. Per-seed reliability diagrams

The 60 figures are organized by transfer pair and architecture. Each figure corresponds to one row in the master results table and provides the visual calibration evidence underlying the ΔECE values reported in the main text. For each direction, ResNet1D and InceptionTime reliability diagrams are shown across seeds 42–46.

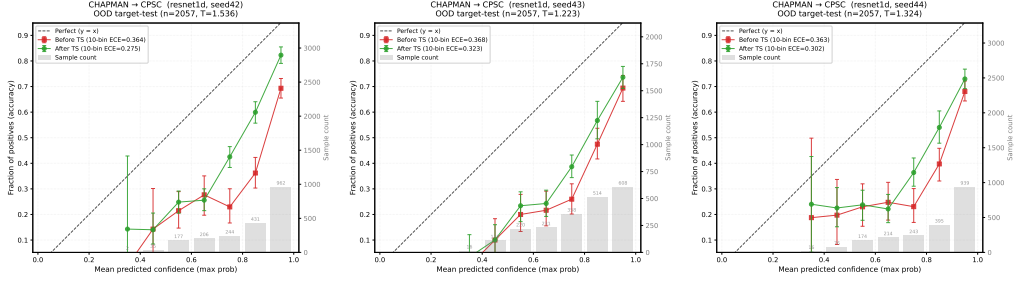


Figure 10: Chapman→CPSC reliability diagrams: ResNet1D seeds 42–44.

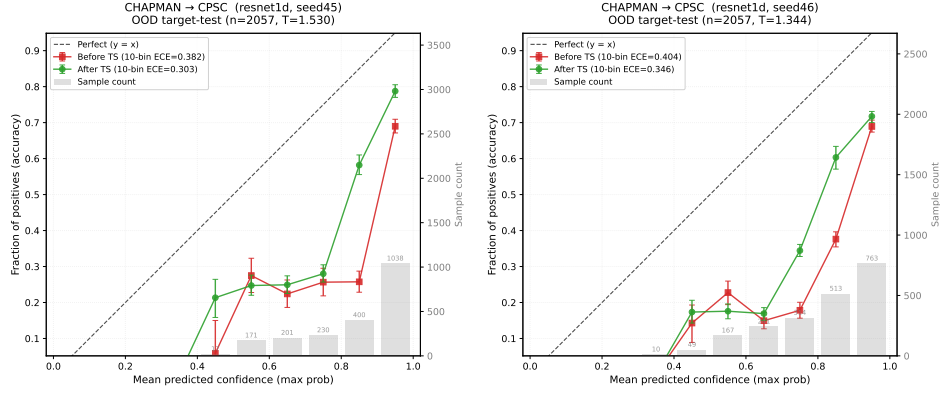


Figure 11: Chapman→CPSC reliability diagrams: ResNet1D seeds 45–46.

S1.1. Chapman→CPSC

S1.2. Chapman→PTB-XL

S1.3. CPSC→Chapman

S1.4. CPSC→PTB-XL

S1.5. PTB-XL→Chapman

S1.6. PTB-XL→CPSC

S2. Additional supplementary materials

- **Per-experiment result files:** full JSON results (checkpoints/transfer/**/transfer_result.json) and CSV summaries (results/*.csv) for all 60 seed experiments.

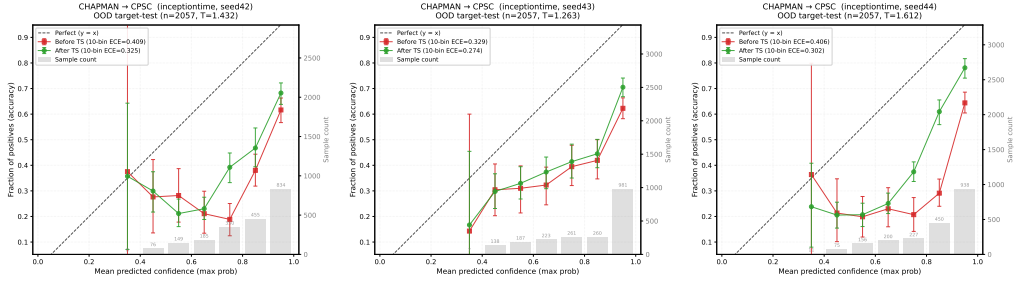


Figure 12: Chapman→CPSC reliability diagrams: InceptionTime seeds 42–44.

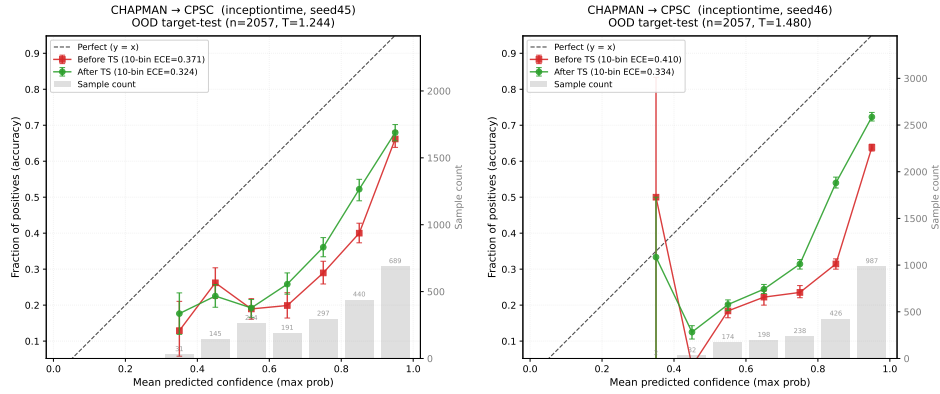


Figure 13: Chapman→CPSC reliability diagrams: InceptionTime seeds 45–46.

- **Pre-registered protocol:** revision A1 full text (docs/PROTOCOL_AMENDMENT_A1_PREREG_RELOCATION.md) and hash manifest (docs/osf_archive_manifest.json).
- **NCV experiment data:** E3 CSV containing `ncv_id` and `ncv_ood` columns for the net clinical value experiment (Section 4.8).
- **Shapley decomposition cells:** 27-cell validation table for the Shapley attribution analysis at $n = 20,000$.

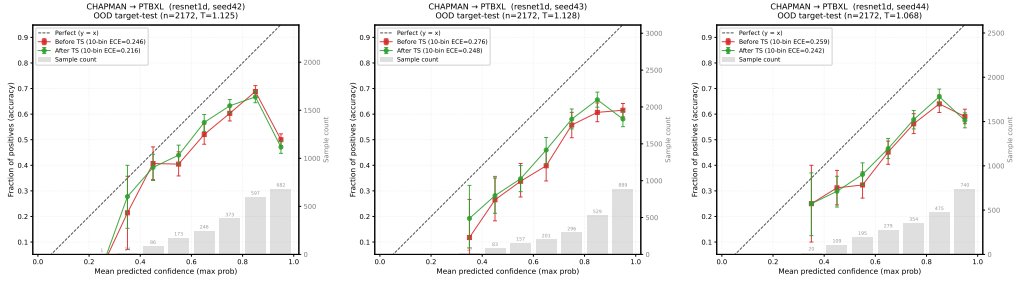


Figure 14: Chapman→PTB-XL reliability diagrams: ResNet1D seeds 42–44.

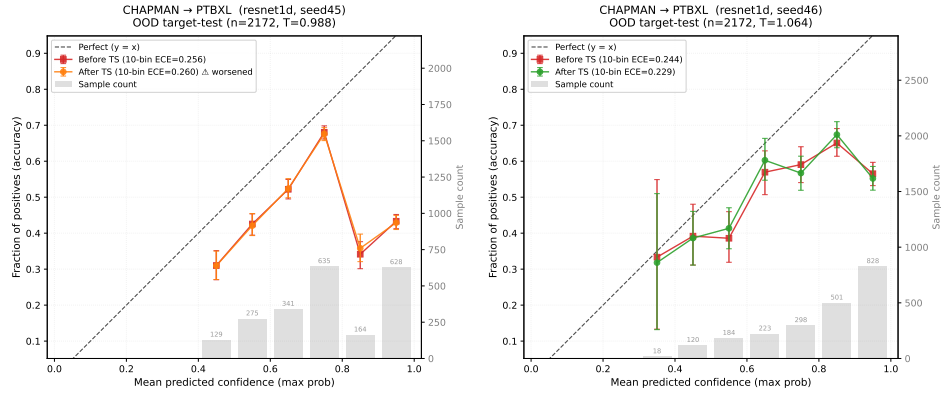


Figure 15: Chapman→PTB-XL reliability diagrams: ResNet1D seeds 45–46.

- **BCa CI rerun manifest:** full BCa bootstrap ($B = 10,000$) CI results for the entire $6 \times 2 \times 5 \times 8$ grid, stored in the `methods.ts.meta` field of each `transfer_result.json`.

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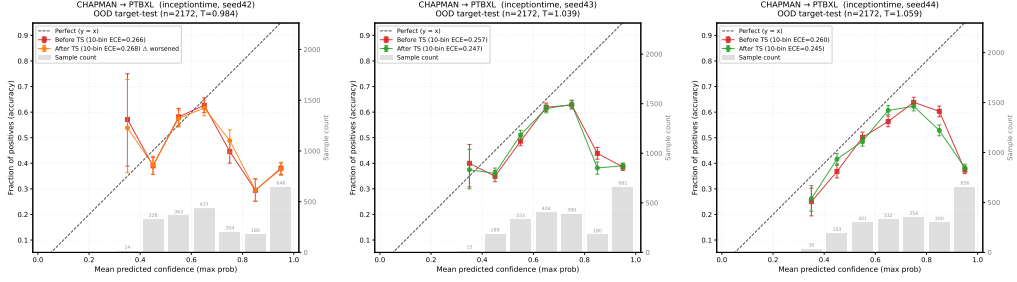


Figure 16: Chapman→PTB-XL reliability diagrams: InceptionTime seeds 42–44.

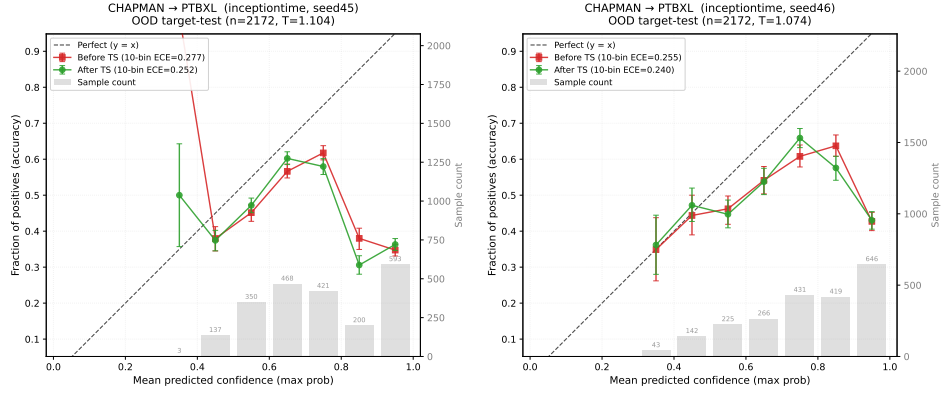


Figure 17: Chapman→PTB-XL reliability diagrams: InceptionTime seeds 45–46.

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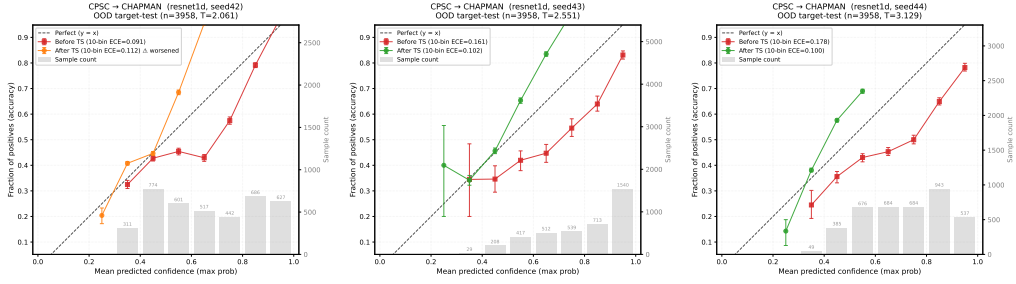


Figure 18: CPSC→Chapman reliability diagrams: ResNet1D seeds 42–44.

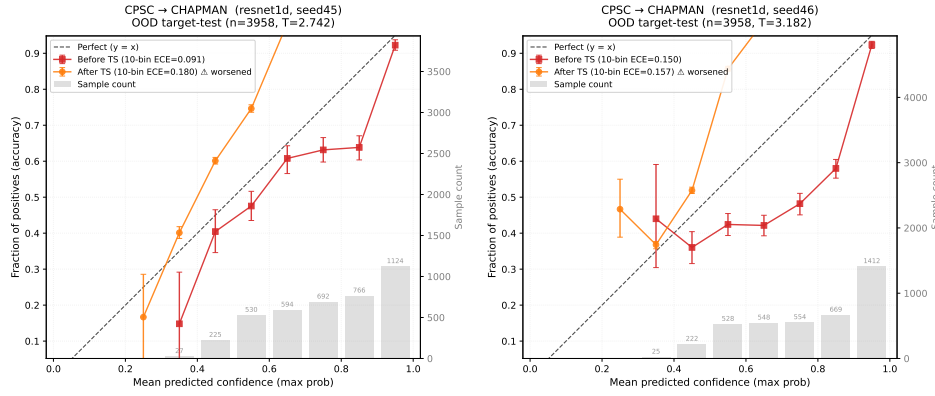


Figure 19: CPSC→Chapman reliability diagrams: ResNet1D seeds 45–46.

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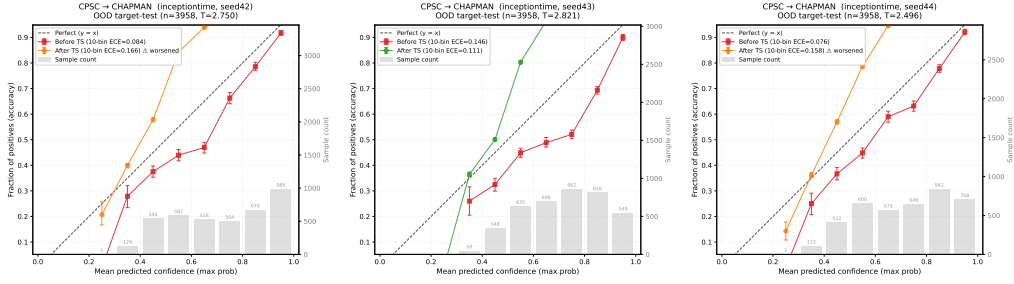


Figure 20: CPSC→Chapman reliability diagrams: InceptionTime seeds 42–44.

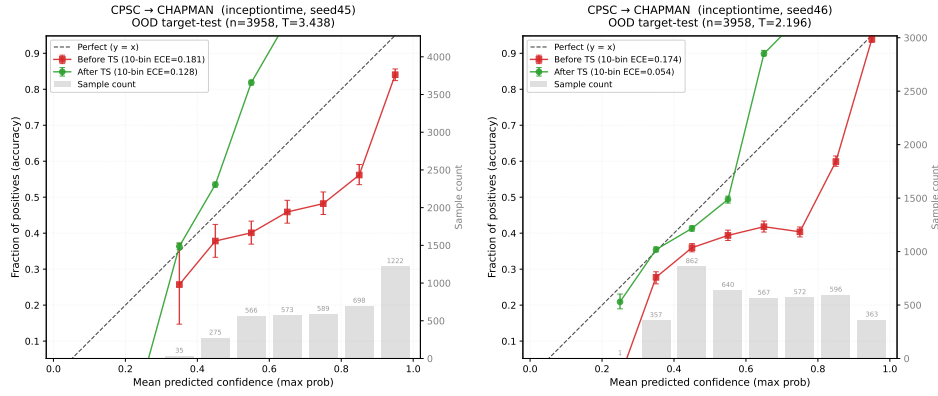


Figure 21: CPSC→Chapman reliability diagrams: InceptionTime seeds 45–46.

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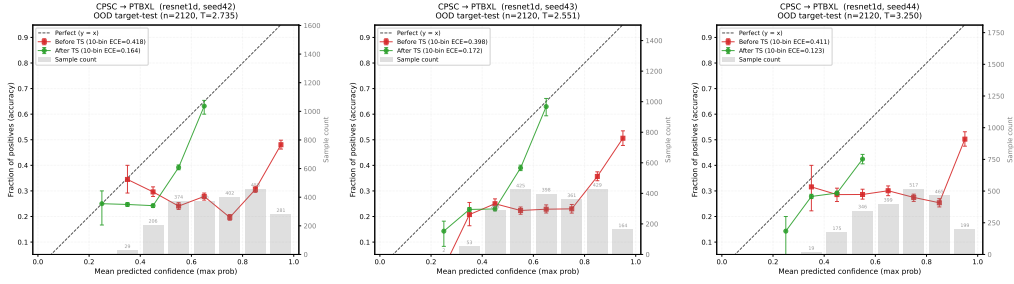


Figure 22: CPSC→PTB-XL reliability diagrams: ResNet1D seeds 42–44.

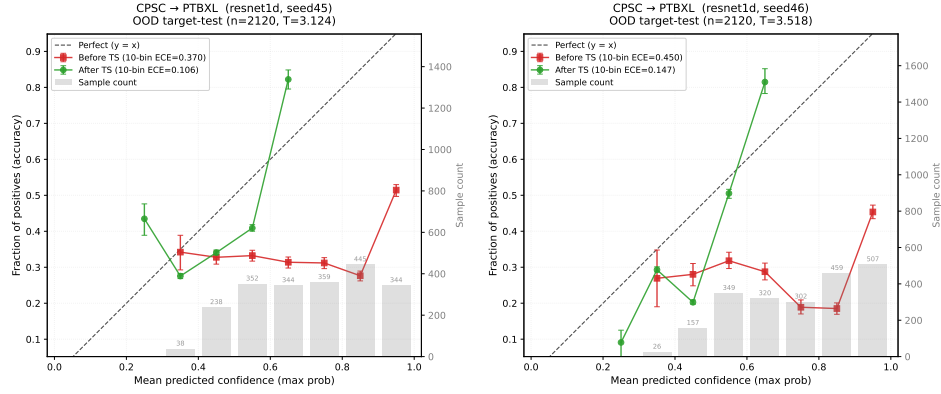


Figure 23: CPSC→PTB-XL reliability diagrams: ResNet1D seeds 45–46.

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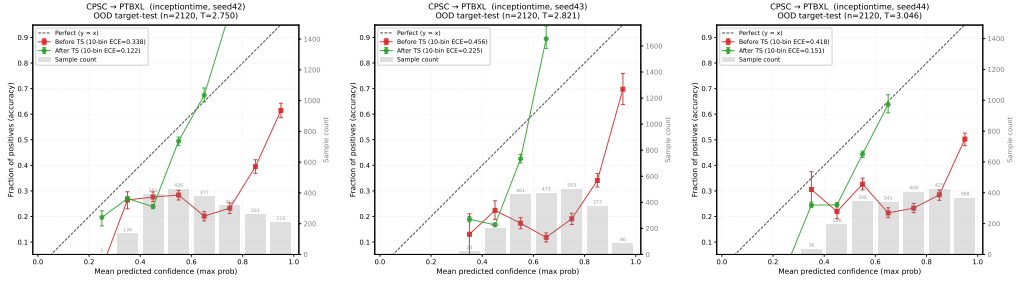


Figure 24: CPSC→PTB-XL reliability diagrams: InceptionTime seeds 42–44.

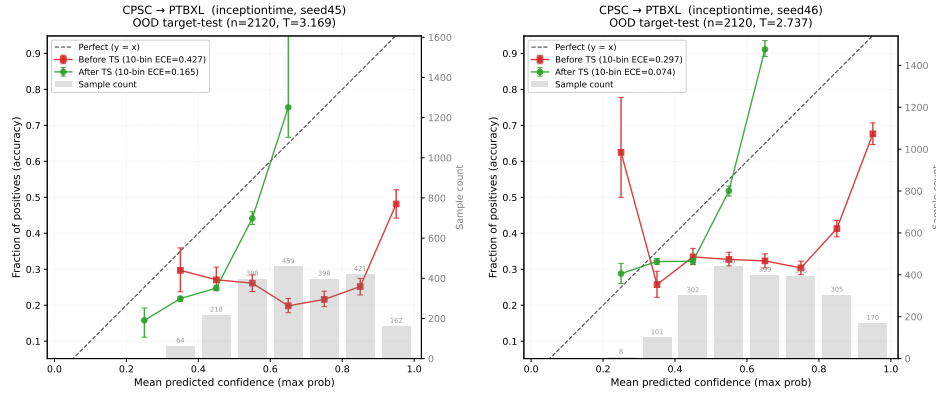


Figure 25: CPSC→PTB-XL reliability diagrams: InceptionTime seeds 45–46.

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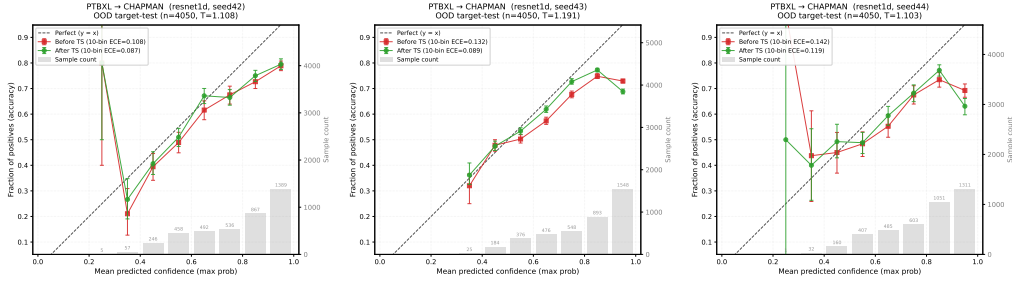


Figure 26: PTB-XL→Chapman reliability diagrams: ResNet1D seeds 42–44.

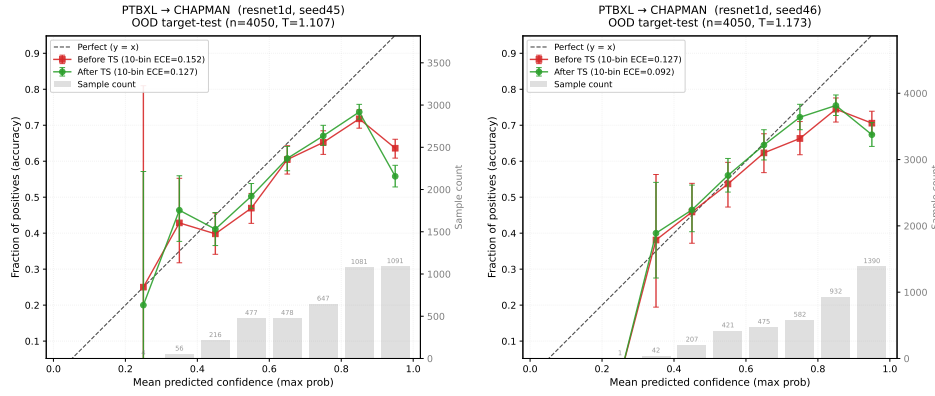


Figure 27: PTB-XL→Chapman reliability diagrams: ResNet1D seeds 45–46.

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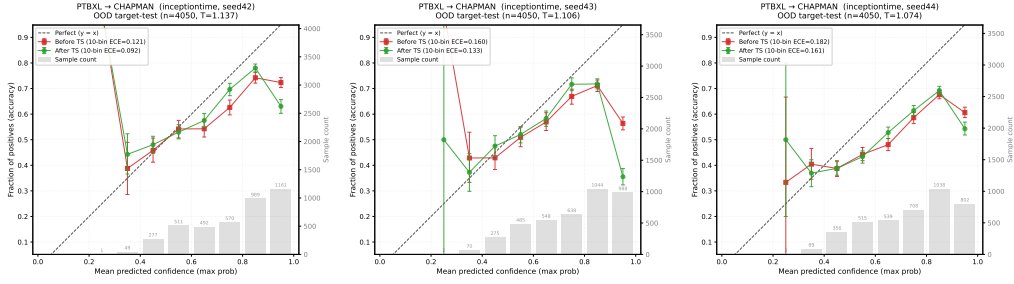


Figure 28: PTB-XL→Chapman reliability diagrams: InceptionTime seeds 42–44.

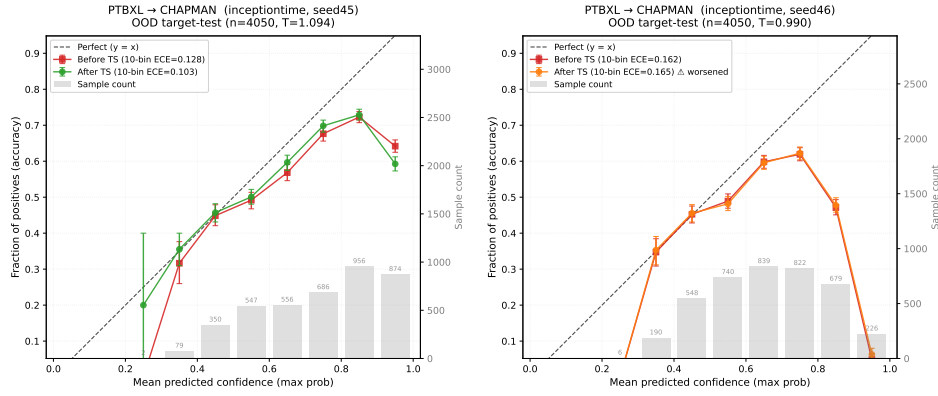


Figure 29: PTB-XL→Chapman reliability diagrams: InceptionTime seeds 45–46.

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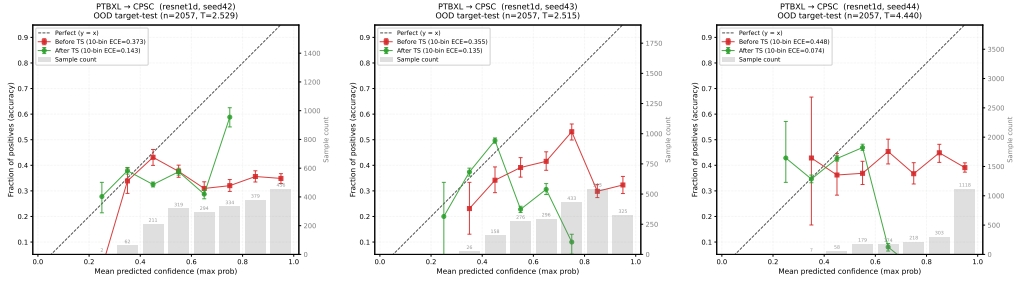


Figure 30: PTB-XL→CPSC reliability diagrams: ResNet1D seeds 42–44.

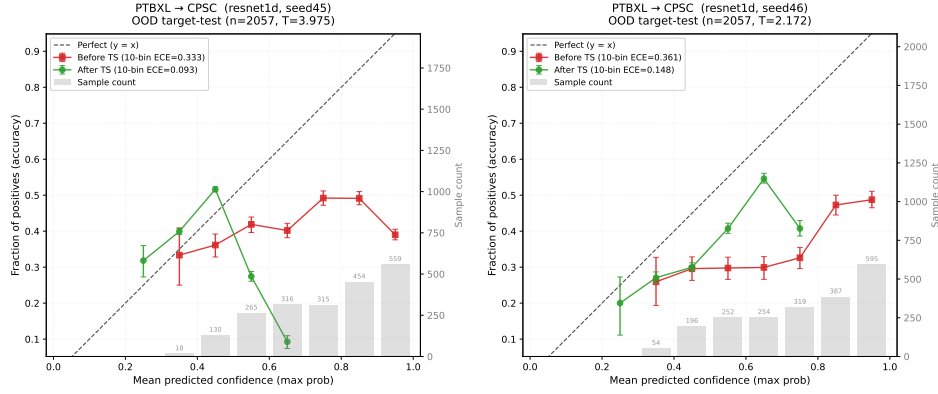


Figure 31: PTB-XL→CPSC reliability diagrams: ResNet1D seeds 45–46.

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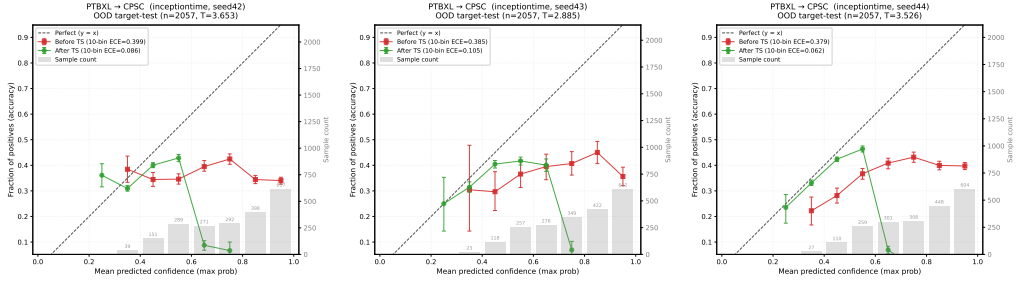


Figure 32: PTB-XL→CPSC reliability diagrams: InceptionTime seeds 42–44.

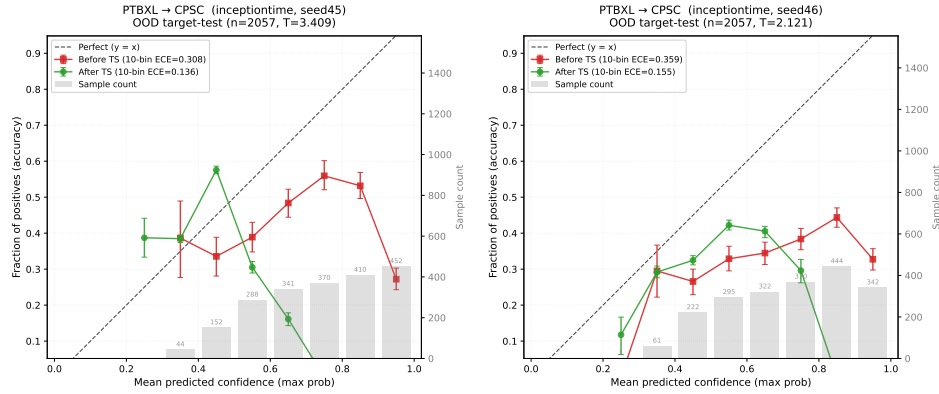


Figure 33: PTB-XL→CPSC reliability diagrams: InceptionTime seeds 45–46.

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Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap→CPSC	Incep	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	ResNet	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap→PTB-XL	Incep	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓
	ResNet	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
CPSC→Chap	Incep	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	ResNet	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
		42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (Inception-Time). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat:* 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 / 0.216 → 0.200	0.217 → 0.201	raw → TS	

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	+0.104	Failure branch
Platt	-0.059	-0.584	+0.002	Failure branch
Isotonic	+0.011	-0.569	+0.101	Failure branch
Vector	-0.061	-0.522	+0.073	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	+0.032	Failure branch
EM prior	+0.143	-0.250	+0.274	Failure branch
BBSE prior	+0.136	-0.354	+0.201	Failure branch

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE $>$ threshold $\rightarrow$ predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155